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(54) **HEREDITARY CANCER GENES**

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CPC **C12Q 1/6886** (2013.01); **A61K 31/138** (2013.01); **C12P 19/34** (2013.01); **G01N 33/57415** (2013.01); **G01N 33/57449** (2013.01); **G16B 20/00** (2019.02); **G16B 20/10** (2019.02); **G16B 20/20** (2019.02); **C12Q 2600/156** (2013.01); **C12Q 2600/16** (2013.01); **G16B 30/00** (2019.02); **G16B 30/10** (2019.02)

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See application file for complete search history.

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ABSTRACT

The invention generally relates to a molecular classification of disease predisposition and particularly to molecular markers for cancer predisposition and methods of use thereof.

9 Claims, 3 Drawing Sheets

Specification includes a Sequence Listing.

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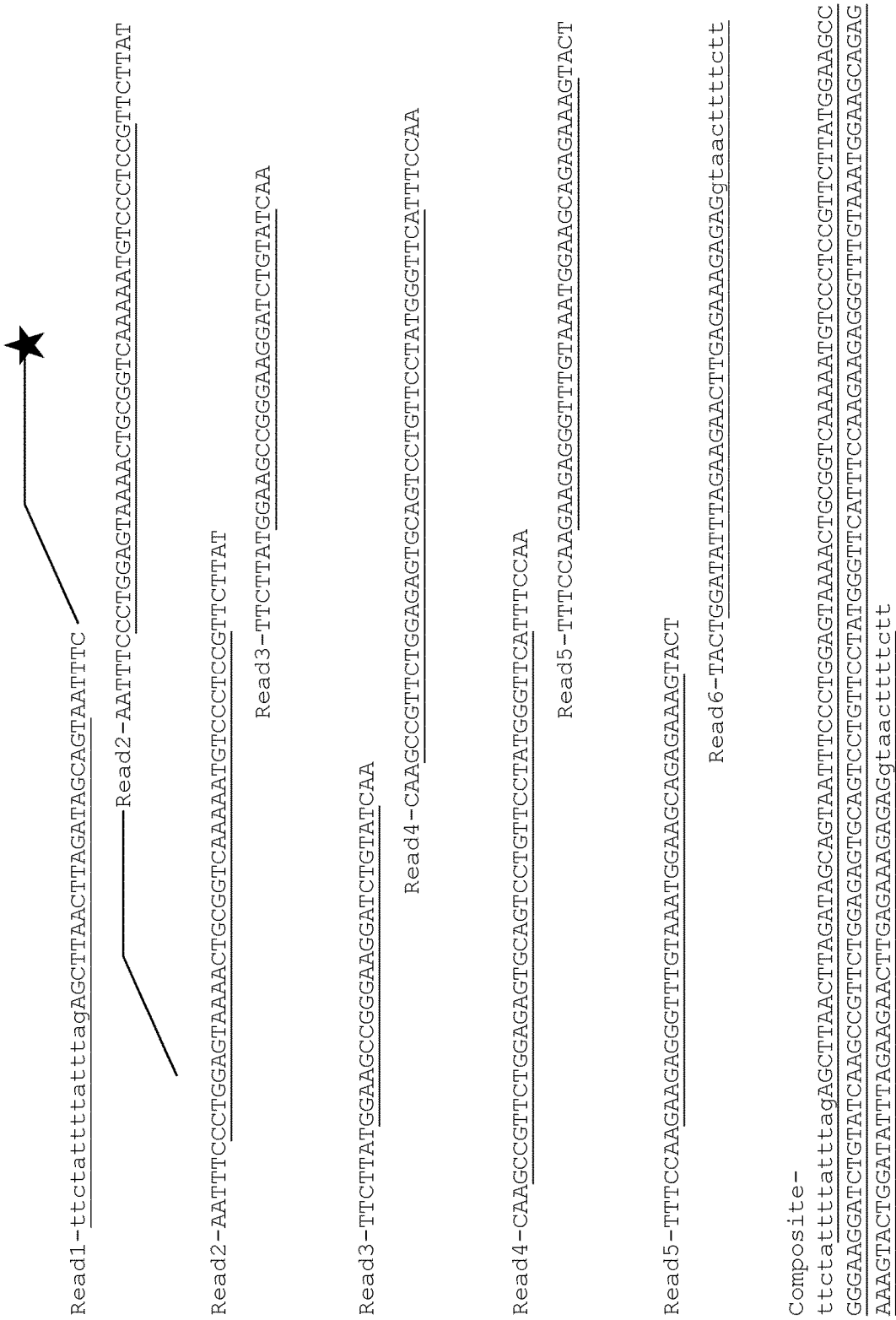


Figure 1

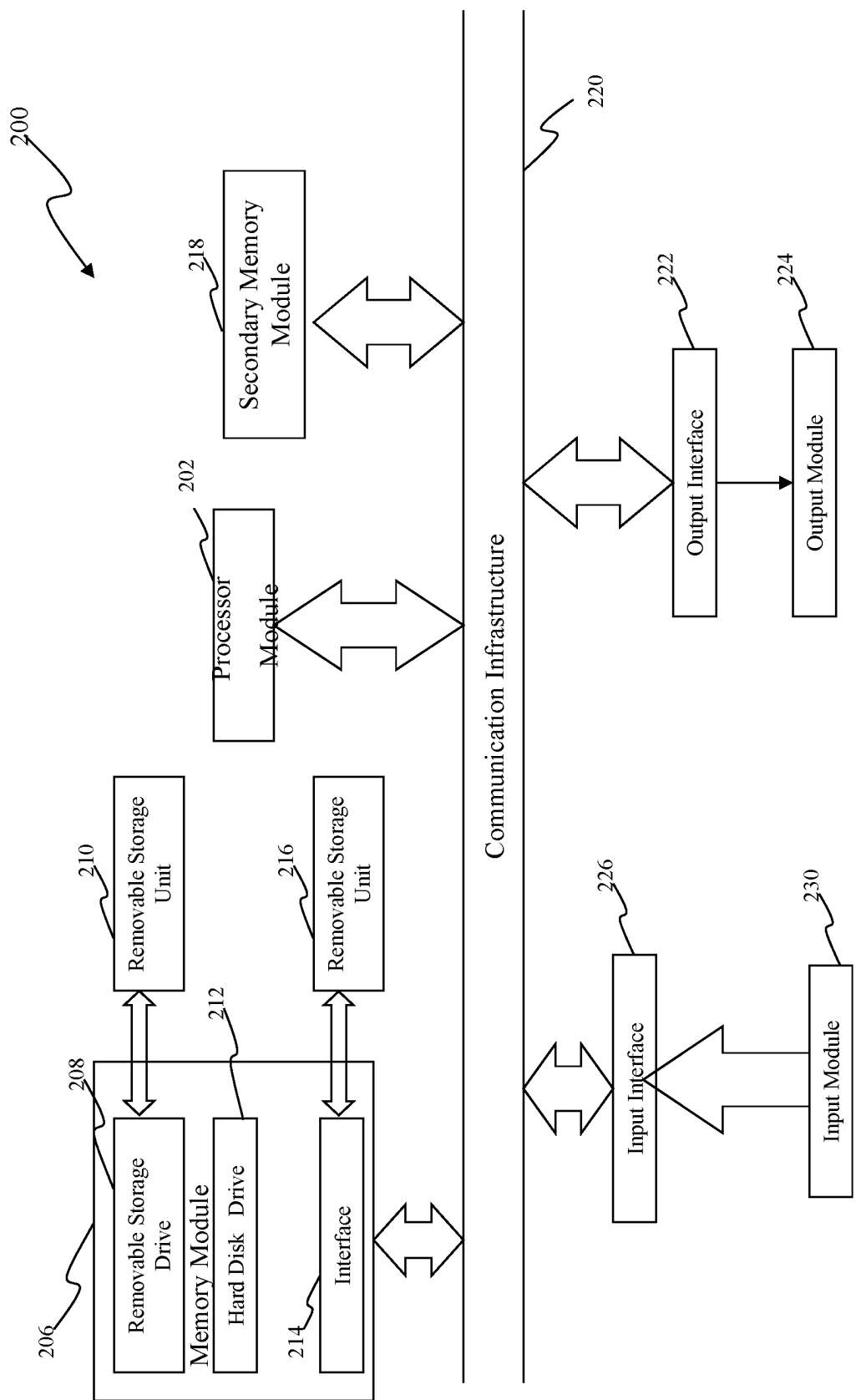


Figure 2

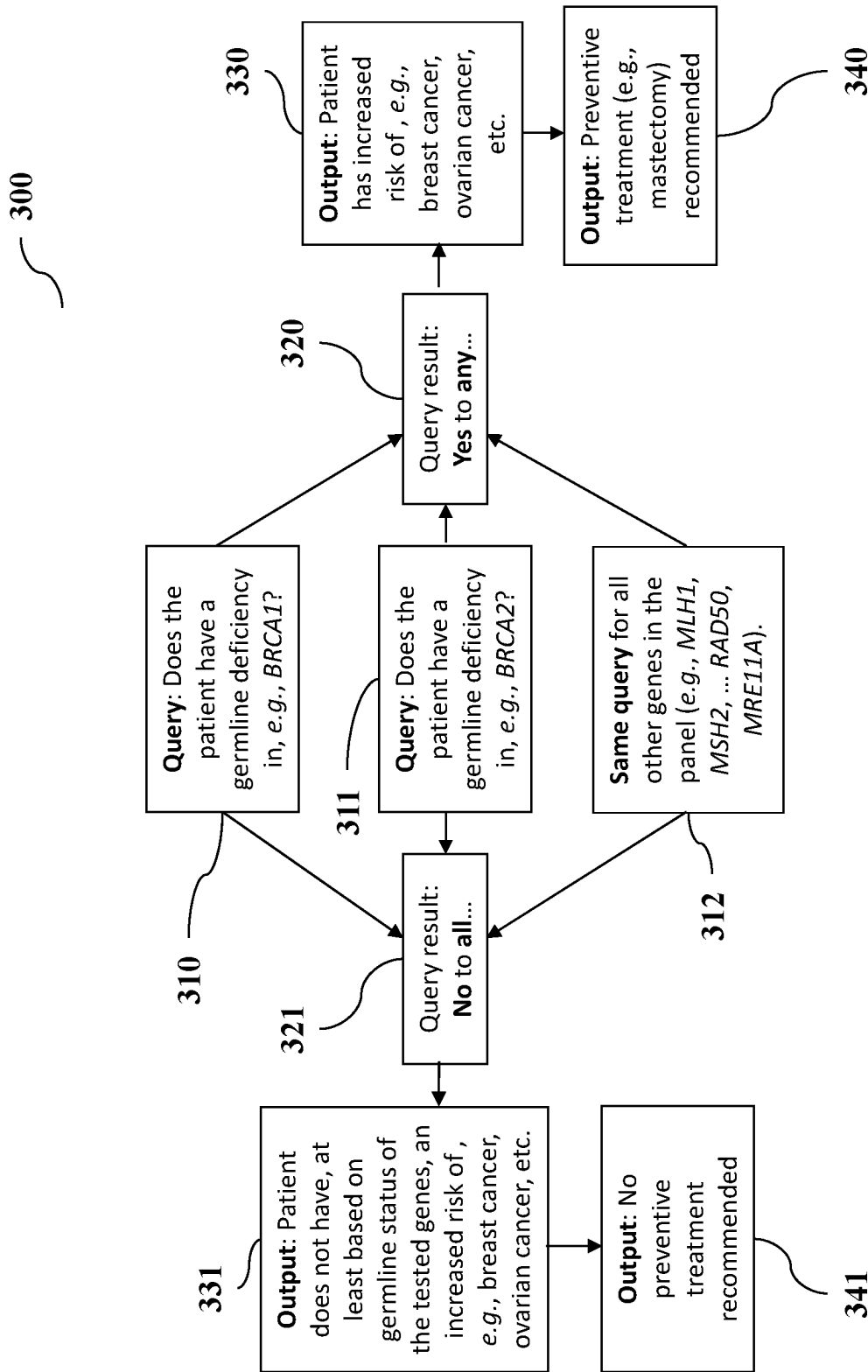


FIGURE 3

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HEREDITARY CANCER GENES**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a divisional of U.S. application Ser. No. 14/561,938, filed Dec. 5, 2014 (issued as U.S. Pat. No. 10,995,338), which is continuation of and claims the priority benefit of Patent Cooperation Treaty International Application Number PCT/US2013/044494 (filed Jun. 6, 2013, published as WO2013184905), which in turn claims the priority benefit of U.S. Provisional Application Ser. Nos. 61/656,333 (filed on Jun. 6, 2012) and 61/814,068 (filed on Apr. 19, 2013), each of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

The invention generally relates to a molecular classification of disease predisposition and particularly to molecular markers for cancer predisposition and methods of use thereof.

SEQUENCE LISTING

The instant application was filed with a formal Sequence Listing submitted electronically as a text file. This text file, which is named "131588-0695_SL.txt", was created on Aug. 22, 2023 and is 1,524,288 bytes in size. Its contents are incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

Cancer is a major public health problem, accounting for roughly 25% of all deaths in the United States. American Cancer Society, *FACTS AND FIGURES 2010*. For many types of cancer, up to 10% of cases can be hereditary. Knowing that a patient has an increased risk of cancer due to hereditary factors can help such a patient to take preventive actions to reduce that risk. Thus, there is a significant need for accurate ways of determining whether a particular patient has an increased risk of cancer.

SUMMARY OF THE INVENTION

The inventors have discovered panels of genes, wherein a deficiency in any of these genes in a patient's germline confers an increased risk of specific cancers. Generally speaking, these genes may be sequenced in patients to determine predisposition to these cancers.

Accordingly, in one aspect the present invention provides a method for sequencing DNA. Generally, the method includes at least the following steps: (1) isolating from a patient sample a plurality of DNA molecules, each DNA molecule consisting of between 50 and 75,000 nucleotides in length, said plurality of DNA molecules comprising one or more exons of a plurality of genes consisting of between 10 and 200 genes, and said plurality of genes comprising APC, BRCA1, BRCA2, CDKN2A, EPCAM, MLH1, MSH2, MSH6, MUTYH, PALB2, and PMS2; and (2) determining the sequence of said plurality of DNA molecules. In some embodiments, the method further comprises comparing the sequences determined in (2) with one or more reference sequences. In some embodiments, the method further comprises correlating a difference between the sequences determined in (2) and the one or more reference sequences to a mutation in one or more of the genes in the

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plurality of genes. In some embodiments the plurality of DNA molecule comprises at least some certain length of intron adjacent to exon (e.g., at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 or more base pairs of the intron).

In another aspect the present invention provides a method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining whether the patient has a germline deficiency in any of a plurality of genes comprising APC, BRCA1, BRCA2, CDKN2A, EPCAM, MLH1, MSH2, MSH6, MUTYH, PALB2, and PMS2; and either (2) correlating a germline deficiency in any of said plurality of genes to an increased risk (e.g., increased hereditary risk) of cancer, or (3) correlating the absence of a germline deficiency in all of said plurality of genes to no increased risk (e.g., no increased hereditary risk) of cancer (or to no identified increased risk due to the tested genes). In some embodiments, the method comprises correlating a germline deficiency in any particular gene in the plurality of genes to an increased risk of a particular cancer as shown in Table 4.

In another aspect the present invention provides a kit comprising: reagents for sequencing DNA molecules comprising one or more exons of a plurality of genes comprising APC, BRCA1, BRCA2, CDKN2A, EPCAM, MLH1, MSH2, MSH6, MUTYH, PALB2, and PMS2; and instructions for using said reagents. In some embodiments the kit comprises reagents for sequencing a plurality of genes consisting of between 11 and 200 genes, and said plurality of genes comprises APC, BRCA1, BRCA2, CDKN2A, EPCAM, MLH1, MSH2, MSH6, MUTYH, PALB2, and PMS2. In some embodiments the reagents are PCR primers specific for the plurality of genes. In some embodiments, the reagents are PCR primers specific for the exons (and optionally some certain amount of adjacent intron) of the plurality of genes.

In some embodiments of the above aspects of the invention, the plurality of genes further comprises at least 1, 2, 3, 4, 5, 6, 7, 8, or 9 genes chosen from the group consisting of ATM, BARD1, BMPR1A, CDH1, CDK4, CHEK2, TP53, PTEN, RAD51D, SMAD4, and STK11. In some embodiments the plurality of genes further comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 genes chosen from the group consisting of BLM, CEBPA, FLCN, MEN1, PTCH, RET, SDH5, SDHB, SDHC, SDHD, TMEM127, and VHL. In some embodiments the plurality of genes further comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 genes chosen from the group consisting of BRAF, BRIP1, FANCA, FANCB, FANCC, FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL, FANCM, KRAS, MLH3, MRE11, NBS1, PIK3CA, PMS1, RAD50, and RAD51C. In some embodiments the plurality of genes further comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, or 54 genes chosen from the group consisting of APC, ATM, BARD1, BLM, BMPR1A, BRAF, BRCA1, BRCA2, BRIP1, CDH1, CDK4, CDKN2A, CEBPA, CHEK2, FANCA, FANCB, FANCC, FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL, FANCM, FLCN, KRAS, MEN1, MLH1, MLH3, MRE11, MSH2, MSH6, MUTYH, NBS1, PALB2, PIK3CA, PMS1, PMS2, PTCH1, PTEN, RAD50, RAD51C, RAD51D, RET, SDHAF2, SDHB, SDHC, SDHD, SMAD4, STK11, EPCAM, TMEM127, TP53, VHL. In some embodiments the plurality of genes further comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,

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23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, or 54 genes of any of Panels A-Q.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention pertains. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

Other features and advantages of the invention will be apparent from the following Detailed Description, and from the Claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates how a plurality of DNA molecules can comprise a particular DNA sequence with no single molecule comprising all of such sequence. Read 1: SEQ ID NO: 603; Read 2: SEQ ID NO: 604; Read 3: SEQ ID NO: 605; Read 4: SEQ ID NO: 606; Read 5: SEQ ID NO: 607; Read 6: SEQ ID NO: 608; Composite: 609.

FIG. 2 is an illustration of an example of a system useful in certain aspects and embodiments of the invention.

FIG. 3 is a flowchart illustrating an example of a computer-implemented method of the invention.

DETAILED DESCRIPTION OF THE INVENTION

The present invention is based in part on the discovery that hereditary cancer genes, and germline deficiencies in these genes, are responsible for increases in cancer risk attributable to heredity. "Hereditary cancer gene" and "HCG" herein refer to a gene wherein germline deficiency in the gene confers an increased risk for cancer. The inventors have discovered specific panels (e.g., pluralities) of HCGs that may be tested in a patient to give a comprehensive understanding of the patient's hereditary cancer risk. All of the HCGs in Table 1 below form a panel of HCGs ("Panel A") useful in the invention.

TABLE 1

Gene #	Entrez Gene Symbol	Entrez Gene ID
1	APC	324
2	ATM	472
3	ATR	545
4	BAP1	8314
5	BARD1	580
6	BLM	641
7	BMPR1A	657
8	BRAF	673
9	BRCA1	672
10	BRCA2	675
11	BRIP1	83990
12	CDH1	999
13	CDK4	1019
14	CDKN2A (p16)	1029
15	CEBPA	1050
16	CFTR	1080
17	CHEK2	11200
18	CTRC	11330
19	EPCAM (TACSTD1)	4072
20	FANCA	2175

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TABLE 1-continued

Gene #	Entrez Gene Symbol	Entrez Gene ID
21	FANCB	2187
22	FANCC	2176
23	FANCD2	2177
24	FANCE	2178
25	FANCF	2188
26	FANCG	2189
27	FANCI	55215
28	FANCL	55120
29	FANCM	57697
30	FGFR2	2263
31	FH	2271
32	FLCN	201163
33	HOXB13	10481
34	HRAS	3265
35	KITLG	4254
36	KRAS	3845
37	MEN1	4221
38	MLH1	4292
39	MLH3	27030
40	MRE11	4361
41	MSH2	4436
42	MSH6	2956
43	MUTYH (MYH)	4595
44	NBS1 (NBN)	4683
45	NF1	4763
46	NF2	4771
47	PALB2	79728
48	PIK3CA	5290
49	PMS1	5378
50	PMS2	5395
51	PRSS1	5644
52	PTCH1	5727
53	PTEN	5728
54	RAD50	10111
55	RAD51C	5889
56	RAD51D	5892
57	RB1	5925
58	RET	5979
59	SDHAF2 (SDH5)	54949
60	SDHB	6390
61	SDHC	6391
62	SDHD	6392
63	SMAD4	4089
64	SPINK1	6690
65	STK11	6794
66	TGFB2	7042
67	TMEM127	55654
68	TP53 (p53)	7157
69	VHL	7428

As will be shown in detail throughout this document, subsets of Panel A can also be used in the invention. Examples of subsets useful in the present invention are shown in Tables 2A to 2D below:

TABLE 2A

Panels B to G						
Gene #	Panel B	Panel C	Panel D	Panel E	Panel F	Panel G
1	BRCA1	BRCA1	BRCA1	MLH1	BRCA1	BRCA1
2	BRCA2	BRCA2	BRCA2	MSH2	BRCA2	BRCA2
3	MLH1	MLH1	CHEK2	MSH6	MLH1	MLH1
4	MSH2	MSH2	ATM	PMS2	MSH2	MSH2
5	MSH6	MSH6	NBN	BRCA1	MSH6	MSH6
6	PMS2	PMS2	PALB2	BRCA2	PMS2	PMS2
7	EPCAM	EPCAM	BARD1	ATM	EPCAM	EPCAM
8	MUTYH	MUTYH	BRIP1	BARD1	APC	APC
9	APC	APC	PMS2	BRIP1	MUTYH	MUTYH
10	CDKN2A	CDKN2A	MSH2	CHEK2	PALB2	PALB2
11	PALB2	PALB2	MSH6	MUTYH	CDKN2A	CHEK2
12	SMAD4	SMAD4	TP53	RAD50	CDK4	PTEN
13	BMPR1A	BMPR1A	MUTYH	EPCAM*	TP53	STK11
14	TP53	TP53			PTEN	CDH1
15	PTEN	PTEN			CDH1	TP53
16	STK11	STK11			STK11	ATM
17	CDH1	CDH1			SMAD4	RAD51C
18	NBN1	NBN1			BMPR1A	RAD51D
19	CHEK2	CHEK2			ATM	BRIP1
20	RAD51C	RAD51C			CHEK2	BARD1
21	RAD51D	RAD51D			RAD51C	BMPR1A
22	BRIP1	BRIP1			RAD51D	SMAD4
23	BARD1	BARD1			MLH3	CDKN2A
24	ATM	ATM			BRIP1	CDK4
25	CDK4	CDK4			BARD1	RAD50
26		RAD50*			NSB	NBN
27		MRE11A*			RAD50	MRE11
28		MLH3*			MRE11A	MLH3
29					HOXB13*	

*Optional

TABLE 2B

Panels H to M						
Gene #	Panel H	Panel I	Panel J	Panel K	Panel L	Panel M
1	APC	ATM	APC	BLM	ATR	BRCA1
2	BRCA1	BMPR1A	ATM	CEBPA	BARD1	BRCA2
3	BRCA2	CDH1	BMPR1A	FLCN	BRAF	MLH1
4	CDKN2A	CDK4	BRCA1	MEN1	BRIP1	MSH2
5	EPCAM	CHEK2	BRCA2	PTCH	FANCA	MSH6
6	MLH1	HOXB13	CDH1	RET	FANCB	PMS2
7	MSH2	TP53	CDK4	SDHAF2	FANCC	EPCAM
8	MSH6	PTEN	CDKN2A	SDHB	FANCD2	MUTYH
9	MUTYH	SMAD4	CHEK2	SDHC	FANCE	APC
10	PALB2	STK11	EPCAM	SDHD	FANCF	CDKN2A
11	PMS2		MLH1	TMEM127	FANCG	PALB2
12			MSH2	VHL	FANCI	SMAD4
13			MSH6		FANCL	BMPR1A
14			MUTYH		FANCM	TP53
15			p53		KRAS	PTEN
16			PALB2		MLH3	STK11
17			PMS2		MRE11	CDH1
18			PTEN		NBS1	NBN1
19			SMAD4		PIK3CA	CHEK2
20			STK11		PMS1	RAD51C
21					RAD50	RAD51D
22					RAD51C	BRIP1
23						BARD1
24						ATM
25						CDK4

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TABLE 2C

Panel N	
Gene #	Gene Symbol
1	BRCA1
2	BRCA2
3	MLH1
4	MSH2
5	PMS2
6	MLH3
7	EPCAM
8	MSH6
9	APC
10	PMS1
11	PTEN
12	STK11
13	RET
14	SDHD
15	SDHC
16	SDHB
17	SDHAF2
18	CDH1
19	MUTYH
20	SMAD4
21	MEN1
22	VHL
23	BMPR1A
24	PALB2
25	TP53
26	FANCL
27	BLM
28	CDK4
29	CDKN2A
30	ATM
31	PTCH1
32	CHEK2
33	RAD51C
34	CEBPA
35	NBS1
36	FANCA
37	FANCC
38	FANCD2
39	FANCE
40	FANCG
41	FANCI
42	FANCM
43	RAD51D
44	FANCF
45	FANCB
46	BARD1
47	RAD50
48	MRE11
49	BRIP1
50	FLCN
51	TMEM127
52	PIK3CA
53	KRAS
54	BRAF
55	HOXB13
56	ATR
57	BAP1
58	CFTR
59	CTRC
60	FGFR2
61	FH
62	HRAS
63	KITLG
64	NF1
65	NF2
66	PRSS1
67	RB1
68	SPINK1
69	TGFB2

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TABLE 2D

Panel O	
Gene #	Gene Symbol
1	BRCA1
2	BRCA2
3	MLH1
4	MSH2
5	MSH6
6	PMS2
7	EPCAM
8	APC
9	MUTYH
10	PALB2
11	CDKN2A
12	CDK4
13	TP53
14	PTEN
15	CDH1
16	STK11
17	SMAD4
18	BMPR1A
19	ATM
20	CHEK2
21	RAD51C
22	RAD51D
23	MLH3
24	VHL
25	MEN1
26	RET
27	NF1
28	NF2
29	RB1
30	PTCH1
31	FH
32	BLM
33	CEBPA
34	FLCN
35	SDHB
36	SDHC
37	SDHD
38	SDHAF2
39	TMEM127
40	CFTR
41	PRSS1
42	CTRC
43	SPINK1
44	KRAS
45	BRIP1
46	BARD1
47	NBS1
48	RAD50
49	FANCA
50	FANCB
51	FANCC
52	FANCD2
53	FANCE
54	FANCF
55	FANCG
56	FANCI
57	FANCL
58	FANCM
59	ATR
60	HRAS
61	TGFB2
62	FGFR2
63	BAP1
64	KITLG
65	BRAF
66	MRE11
67	PIK3CA
68	PMS1
69	HOXB13

Aspects of the Invention

Accordingly, in one aspect the present invention provides a method for sequencing nucleic acids. Generally, the

method includes at least the following steps: (1) isolating a plurality of nucleic acid molecules from a sample taken from a patient, each nucleic acid molecule comprising (or consisting of or consisting essentially of) between A and B nucleotides in length, said plurality of nucleic acid molecules comprising (e.g., having nucleotide sequences that together comprise) one or more exons of a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q; and (2) determining the sequence of said plurality of nucleic acid molecules.

In another aspect the present invention provides a method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; and either (2)(a) correlating a germline deficiency in any of said plurality of genes to an increased risk (e.g., increased hereditary risk) of cancer, or (2)(b) correlating the absence of a germline deficiency in all of said plurality of genes to no increased risk (e.g., no increased hereditary risk) of cancer. In some embodiments of this aspect, the method also comprises (a) isolating a plurality of nucleic acid molecules from a sample taken from a patient, each nucleic acid molecule comprising (or consisting of or consisting essentially of) between A and B nucleotides in length, and said plurality of nucleic acid molecules comprising (e.g., having nucleotide sequences that together comprise) one or more exons of said plurality of genes and (b) determining the sequence of said plurality of nucleic acid molecules. In some embodiments, the method comprises detecting a germline deficiency in a gene by comparing the sequence determined in (b) with one or more reference sequences, as discussed in more detail below.

Thus, the invention provides a method treating a patient comprising (1) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; and (2)(a) correlating a germline deficiency in any of said plurality of genes to an increased risk (e.g., increased hereditary risk) of cancer (e.g., a particular cancer), or (2)(b) correlating the absence of a germline deficiency in all of said plurality of genes to no increased risk (e.g., no increased hereditary risk) of cancer; and (3) recommending, prescribing, or administering a treatment to reduce the patient's risk of cancer. In some embodiments, the treatment comprises removing all or part of the organ in which the patient has an increased risk of cancer (e.g., mastectomy, salpingo-oophorectomy, hysterectomy, colectomy, prostatectomy, etc.). In some embodiments the treatment comprises preventive drug treatments (e.g., tamoxifen treatment in patients with increased risk of breast or ovarian cancer).

Another aspect of the present invention provides computer program products comprising a computer-usable medium having computer-readable program codes or instructions embodied thereon for enabling a processor to carry out the methods of the invention. A related aspect of the present invention provides a system for determining whether a patient has increased likelihood of response to a particular treatment regimen. Generally speaking, the system comprises (1) computer program for receiving, storing, and/or retrieving a patient's sequence data for a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; (2) computer program for querying this patient data; (3)

optionally a computer program for comparing the patient's sequence data to one or more reference sequences to determine whether there is a mutation; (4) computer program for concluding whether there is an increased likelihood of cancer based on the presence or absence of a mutation; and optionally (4) computer program for outputting/displaying this conclusion. In some embodiments this program for outputting the conclusion may comprise a computer program for informing a health care professional of the conclusion.

In another aspect the invention provides a system for sequencing genes in a sample (e.g., tumor sample), comprising: (1) a sample analyzer for sequencing a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q, wherein the sample analyzer contains (a) the sample which is from a patient, (b) genomic DNA from the sample, (c) transcript RNA from the sample, or (d) DNA synthesized from said genomic DNA; (2) a first computer program for receiving test sequence data on the plurality of genes; and (3) a second computer program for comparing the sequence data to one or more reference sequences. In some embodiments the system comprises a computer program for determining (including quantifying) the patient's degree of risk of cancer based at least in part on the comparison of the test sequence with said one or more reference sequences. Such program may also compare the patient's determined probability of a particular cancer with a reference probability to determine whether the patient has an increased risk of such cancer.

In another aspect the invention provides methods combining the genetic analysis as described above with analysis of other cancer risk factors, e.g., a patient's family and/or personal history of cancer. In some embodiments the invention provides a method for determining a patient's risk of cancer, which comprises: (1)(a) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q and (1)(b) assigning a first risk level of cancer (e.g., percentage probability of developing cancer (any cancer or a specific cancer or set of cancers) by a certain age) for the patient based on the presence or absence of such germline deficiency; (2)(a) evaluating the patient's personal and family history risk factors for cancer and (2)(b) assigning a second risk level of cancer for the patient based on the risk factors identified in (2)(a); and either (3)(a) assigning (optionally communicating and/or recording) the higher of the first and second risk levels determined in (1)(b) and (2)(b) to the patient, or (3)(b) assigning (optionally communicating and/or recording) a third risk level of cancer to the patient, wherein the third risk level is a combination of the first and second risk levels determined in (1)(b) and (2)(b). In some embodiments, the first and second risk levels are given approximately the same weight (e.g., within 5% or 10%) in assigning the third risk level. In some embodiments the ratio of the weight given to the first level to the weight given to the second risk level is approximately 1:2, 1:3, 1:4, 1:5, 1:6, 1:7, 1:8, 1:9, 1:10, 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 2:3, 3:4, 4:5, 5:6, 6:7, 7:8, 8:9, 9:10, 10:11, 3:2, 4:3, 5:4, 6:5, 7:6, 8:7, 9:8, 10:9, 11:10, 3:5, 5:7, 7:9, 9:11, 11:9, 9:7, 7:5, or 5:3. In some embodiments, both the first risk level and the second risk level are communicated (e.g., to the healthcare provider, to the patient, etc.). Personal risk factors may include cancer diagnosis (including age at diagnosis), multiple primary cancers, triple negative breast cancer, ovarian cancer, smoking, age of menopause, age of menarche,

positive biopsy, positive pap smear, male breast cancer, enlarged prostate, colon polyps, etc. Family risk factors can include a relative (e.g., first or second degree) with early onset (e.g., before 40, 50, or 60 years of age) cancer, particular ancestries (e.g., Ashkenazi Jewish ancestry), relative with multiple primary cancers, relative with male breast cancer, relative with ovarian cancer, relative with triple negative breast cancer, etc.

In another aspect the invention provides compositions for use in the above methods. Such compositions include, but are not limited to: (a) nucleic acid probes hybridizing to a plurality of nucleic acid molecules comprising (e.g., having nucleotide sequences that together comprise) one or more exons of a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q; (b) nucleic acid primers and primer pairs suitable for selectively amplifying nucleic acids of (a); (c) antibodies binding immunologically to polypeptides encoded by a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q; (d) a probe set comprising (a), (b) and/or (c); (e) a microarray comprising (a), (b), (c), and/or (d).

In another aspect the present invention provides a kit comprising: reagents for sequencing nucleic acid molecules comprising one or more exons of a plurality of genes comprising a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; and instructions for using said reagents. In some embodiments the kit comprises (a), (b), (c), (d), and/or (e) in the preceding paragraph. In some embodiments the reagents are PCR primers specific for the plurality of genes. In some embodiments, the reagents are PCR primers specific for the exons (and optionally some certain amount of adjacent intron) of the plurality of genes (optionally also including polymerase enzyme, deoxynucleotides, buffers, etc.). In some embodiments, the reagents are oligonucleotide probes specific for the exons (and optionally some certain amount of adjacent intron) of the plurality of genes. In some embodiments the reagents (e.g., the primers and/or probes) are packaged into an array (e.g., affixed to a solid support, contained within a reaction volume, etc.).

Several aspects of the invention described herein involve a step of correlating a particular assay or analysis result or output (e.g., presence or absence of a germline deficiency in one or more genes of Panel B) to some likelihood (e.g., increased, not increased, decreased, etc.) of some clinical feature (e.g., increased risk (e.g., increased hereditary risk) of cancer). Throughout this document, wherever such an aspect is described, an alternative aspect of the invention may involve, in addition to or instead of a correlating step, one or both of the following steps: (a) concluding that the patient has or does not have the clinical feature based at least in part on the assay or analysis result; or (b) communicating that the patient has or does not have the clinical feature based at least in part on the assay or analysis result.

By way of illustration, but not limitation, one aspect described in this document is a method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; and either (2)(a) correlating a germline deficiency in any of said plurality of genes to an increased risk (e.g., increased hereditary risk) of cancer, or (2)(b) correlating the absence of a germline deficiency in all of said plurality of genes to no increased

risk (e.g., no increased hereditary risk) of cancer (or to no identified increased risk due to the tested genes). According to the preceding paragraph, this description of this aspect is understood to include a description of two alternative related aspects. One such embodiment provides a method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two test genes in any of Panels A-Q; and either (2)(a) concluding the patient an increased risk (e.g., increased hereditary risk) of cancer based at least in part on the presence of a germline deficiency in any of said plurality of genes (or in any of said test genes); or (2)(b) concluding the patient does not have an increased risk (e.g., no increased hereditary risk) of cancer based at least in part on the absence of a germline deficiency in each of said plurality of genes (or in each of said test genes) (or alternatively concluding the patient has no identified increased risk due to the tested genes). Another such embodiment provides a method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining whether the patient has a germline deficiency in any of a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two test genes in any of Panels A-Q; and either (2)(a) communicating (e.g., reporting) that the patient an increased risk (e.g., increased hereditary risk) of cancer based at least in part on the presence of a germline deficiency in any of said plurality of genes (or in any of said test genes); or (2)(b) communicating (e.g., reporting) that the patient does not have an increased risk (e.g., no increased hereditary risk) of cancer based at least in part on the absence of a germline deficiency in each of said plurality of genes (or in each of said test genes) (or alternatively communicating that the patient has no identified increased risk due to the tested genes).

In each embodiment described in this document involving correlating a particular assay or analysis result or output (e.g., presence or absence of a germline deficiency in one or more genes of Panel B) to some likelihood (e.g., increased, not increased, decreased, etc.) of some clinical feature (e.g., increased risk (e.g., increased hereditary risk) of cancer), or additionally or alternatively concluding or communicating such clinical feature based at least in part on such particular assay or analysis result, such correlating, concluding or communicating may comprise assigning a risk or likelihood of the clinical feature occurring based at least in part on the particular assay or analysis result. In some embodiments, such risk is a percentage probability of the event or outcome occurring. In some embodiments, the patient is assigned to a risk group (e.g., low risk, intermediate risk, high risk, etc.). In some embodiments "low risk" is any percentage probability below 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or 50%. In some embodiments "intermediate risk" is any percentage probability above 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or 50% and below 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, or 75%. In some embodiments "high risk" is any percentage probability above 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99%.

As used herein, "communicating" a particular piece of information means to make such information known to another person or transfer such information to a thing (e.g., a computer). In some methods of the invention, a patient's qualitative or quantitative risk of cancer (e.g., a specific cancer or syndrome listed in Table 4) is communicated. In some embodiments, the information used to arrive at such a

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risk prediction (e.g., presence or absence of germline deficiency in one or more genes in Panel B) is communicated. This communication may be auditory (e.g., verbal), visual (e.g., written), electronic (e.g., data transferred from one computer system to another), etc. In some embodiments, communicating a cancer risk (e.g., "increased", "not increased", "up to X %", etc.) comprises generating a report that communicates the risk. In some embodiments the report is a paper report, an auditory report, or an electronic record. In some embodiments the report is displayed and/or stored on a computing device (e.g., handheld device, desktop computer, smart device, website, etc.). In some embodiments the cancer risk is communicated to a physician (e.g., a report communicating the risk is provided to the physician). In some embodiments the cancer risk is communicated to a patient (e.g., a report communicating the risk is provided to the patient). Communicating a cancer risk can also be accomplished by transferring information (e.g., data) embodying the risk to a server computer and allowing an intermediary or end-user to access such information (e.g., by viewing the information as displayed from the server, by downloading the information in the form of one or more files transferred from the server to the intermediary or end-user's device, etc.).

Wherever an embodiment of the invention comprises concluding some clinical feature (e.g., increased risk of cancer, etc.), this may include in some embodiments a computer program concluding such feature, typically after performing an algorithm that applies information on germline deficiency in HCGs according to the present invention.

EMBODIMENTS OF THESE ASPECTS

Various embodiments of the preceding aspects of the invention are provided. Unless otherwise stated, the invention may apply each of these embodiments to each of the preceding aspects.

In some embodiments, the method or system comprises comparing the sequences determined in an earlier step or other computer program with one or more reference sequences. In some embodiments, the method comprises correlating a difference between the determined sequences and the one or more reference sequences to a mutation in one or more of the genes in the plurality of genes. In some embodiments the system comprises a computer program for determining whether the patient has a mutation in one or more of the genes in the plurality of genes by determining whether there is a difference between the determined sequences and the one or more reference sequences. In some embodiments the reference sequence for any given gene in the panel is any of the sequences corresponding to that gene as shown in Table 3 below:

TABLE 3

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
1	APC	NM_001127511.2	trans-var-1
2	APC	NM_001127510.2	trans-var-2
3	APC	NM_000038.5	trans-var-3
4	APC		
5	APC	Exon 1	501-878
6	APC	Exon 2	501-585
7	APC	Exon 3	501-702
8	APC	Exon 4	501-609

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
9	APC	Exon 5	501-614
10	APC	Exon 6	501-605
11	APC	Exon 7	501-599
12	APC	Exon 8	501-879
13	APC	Exon 9	501-596
14	APC	Exon 10	501-640
15	APC	Exon 11	501-578
16	APC	Exon 12	501-617
17	APC	Exon 13	501-715
18	APC	Exon 14	501-9187
19	APC		
20	ATM	NM_000051.3	
21	BARD1	NM_000465.2	
22	BARD1		
23	BARD1	Exon 1	501-793
24	BARD1	Exon 2	501-557
25	BARD1	Exon 3	501-649
26	BARD1	Exon 4	501-1450
27	BARD1	Exon 5	501-581
28	BARD1	Exon 6	501-673
29	BARD1	Exon 7	501-609
30	BARD1	Exon 8	501-633
31	BARD1	Exon 9	501-593
32	BARD1	Exon 10	501-598
33	BARD1	Exon 11	501-958
34	BARD1		
35	BLM	NM_000057.2	
36	BLM		
37	BLM	Exon 1	501-593
38	BLM	Exon 2	501-602
39	BLM	Exon 3	501-1201
40	BLM	Exon 4	501-660
41	BLM	Exon 5	501-628
42	BLM	Exon 6	501-633
43	BLM	Exon 7	501-1162
44	BLM	Exon 8	501-692
45	BLM	Exon 9	501-619
46	BLM	Exon 10	501-614
47	BLM	Exon 11	501-599
48	BLM	Exon 12	501-649
49	BLM	Exon 13	501-607
50	BLM	Exon 14	501-661
51	BLM	Exon 15	501-696
52	BLM	Exon 16	501-691
53	BLM	Exon 17	501-648
54	BLM	Exon 18	501-700
55	BLM	Exon 19	501-693
56	BLM	Exon 20	501-623
57	BLM	Exon 21	501-702
58	BLM	Exon 22	501-855
59	BLM		
60	BMPRI1A	NM_004329.2	
61	BMPRI1A		
62	BMPRI1A	Exon 1	501-781
63	BMPRI1A	Exon 2	501-615
64	BMPRI1A	Exon 3	501-719
65	BMPRI1A	Exon 4	501-663
66	BMPRI1A	Exon 5	501-603
67	BMPRI1A	Exon 6	501-597
68	BMPRI1A	Exon 7	501-600
69	BMPRI1A	Exon 8	501-645
70	BMPRI1A	Exon 9	501-693
71	BMPRI1A	Exon 10	501-798
72	BMPRI1A	Exon 11	501-676
73	BMPRI1A	Exon 12	501-631
74	BMPRI1A	Exon 13	501-2095
75	BMPRI1A		
76	BRAF	NM_004333.4	
77	BRAF		
78	BRAF	Exon 1	501-699
79	BRAF	Exon 2	501-602
80	BRAF	Exon 3	501-764
81	BRAF	Exon 4	501-604
82	BRAF	Exon 5	501-603

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
83	BRAF	Exon 6	501-649
84	BRAF	Exon 7	501-620
85	BRAF	Exon 8	501-660
86	BRAF	Exon 9	501-537
87	BRAF	Exon 10	501-637
88	BRAF	Exon 11	501-618
89	BRAF	Exon 12	501-585
90	BRAF	Exon 13	501-677
91	BRAF	Exon 14	501-547
92	BRAF	Exon 15	501-619
93	BRAF	Exon 16	501-632
94	BRAF	Exon 17	501-635
95	BRAF	Exon 18	501-1258
96	BRAF		
97	BRCA1	NM_007294.3	trans-var-1
98	BRCA1	NM_007300.3	trans-var-2
99	BRCA1	NM_007297.3	trans-var-3
100	BRCA1	NM_007298.3	trans-var-4
101	BRCA1	NM_007299.3	trans-var-5
102	BRCA1		
103	BRCA1	Exon 1	501-713
104	BRCA1	Exon 2	501-599
105	BRCA1	Exon 3	501-554
106	BRCA1	Exon 4	501-578
107	BRCA1	Exon 5	501-589
108	BRCA1	Exon 6	501-640
109	BRCA1	Exon 7	501-606
110	BRCA1	Exon 8	501-546
111	BRCA1	Exon 9	501-577
112	BRCA1	Exon 10	501-3926
113	BRCA1	Exon 11	501-589
114	BRCA1	Exon 12	501-672
115	BRCA1	Exon 13	501-566
116	BRCA1	Exon 14	501-624
117	BRCA1	Exon 15	501-691
118	BRCA1	Exon 16	501-811
119	BRCA1	Exon 17	501-588
120	BRCA1	Exon 18	501-578
121	BRCA1	Exon 19	501-541
122	BRCA1	Exon 20	501-584
123	BRCA1	Exon 21	501-555
124	BRCA1	Exon 22	501-574
125	BRCA1	Exon 23	501-561
126	BRCA1	Exon 24	501-2008
127	BRCA1		
128	BRCA2	NM_000059.3	
129	BRCA2		
130	BRCA2	Exon 1	501-688
131	BRCA2	Exon 2	501-606
132	BRCA2	Exon 3	501-749
133	BRCA2	Exon 4	501-609
134	BRCA2	Exon 5	501-550
135	BRCA2	Exon 6	501-541
136	BRCA2	Exon 7	501-615
137	BRCA2	Exon 8	501-550
138	BRCA2	Exon 9	501-612
139	BRCA2	Exon 10	501-1616
140	BRCA2	Exon 11	501-5432
141	BRCA2	Exon 12	501-596
142	BRCA2	Exon 13	501-570
143	BRCA2	Exon 14	501-928
144	BRCA2	Exon 15	501-682
145	BRCA2	Exon 16	501-688
146	BRCA2	Exon 17	501-671
147	BRCA2	Exon 18	501-855
148	BRCA2	Exon 19	501-656
149	BRCA2	Exon 20	501-645
150	BRCA2	Exon 21	501-622
151	BRCA2	Exon 22	501-699
152	BRCA2	Exon 23	501-664
153	BRCA2	Exon 24	501-639
154	BRCA2	Exon 25	501-745
155	BRCA2	Exon 26	501-647
156	BRCA2	Exon 27	501-2011

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
157	BRCA2		
158	BRIP1	NM_032043.2	
159	BRIP1	NM_032043.2	
160	BRIP1		
161	BRIP1	Exon 1	501-776
162	BRIP1	Exon 2	501-623
163	BRIP1	Exon 3	501-612
164	BRIP1	Exon 4	501-674
165	BRIP1	Exon 5	501-628
166	BRIP1	Exon 6	501-620
167	BRIP1	Exon 7	501-791
168	BRIP1	Exon 8	501-722
169	BRIP1	Exon 9	501-700
170	BRIP1	Exon 10	501-633
171	BRIP1	Exon 11	501-655
172	BRIP1	Exon 12	501-666
173	BRIP1	Exon 13	501-641
174	BRIP1	Exon 14	501-662
175	BRIP1	Exon 15	501-660
176	BRIP1	Exon 16	501-622
177	BRIP1	Exon 17	501-613
178	BRIP1	Exon 18	501-583
179	BRIP1	Exon 19	501-830
180	BRIP1	Exon 20	501-5455
181	BRIP1		
182	CDH1	NM_004360.3	
183	CDH1		
184	CDH1	Exon 1	501-672
185	CDH1	Exon 2	501-615
186	CDH1	Exon 3	501-724
187	CDH1	Exon 4	501-644
188	CDH1	Exon 5	501-656
189	CDH1	Exon 6	501-645
190	CDH1	Exon 7	501-676
191	CDH1	Exon 8	501-629
192	CDH1	Exon 9	501-683
193	CDH1	Exon 10	501-745
194	CDH1	Exon 11	501-646
195	CDH1	Exon 12	501-725
196	CDH1	Exon 13	501-728
197	CDH1	Exon 14	501-631
198	CDH1	Exon 15	501-644
199	CDH1	Exon 16	501-2752
200	CDH1		
201	CDK4	NM_000075.3	
202	CDK4		
203	CDK4	Exon 1	501-773
204	CDK4	Exon 2	501-737
205	CDK4	Exon 3	501-636
206	CDK4	Exon 4	501-668
207	CDK4	Exon 5	501-610
208	CDK4	Exon 6	501-551
209	CDK4	Exon 7	501-636
210	CDK4	Exon 8	501-1391
211	CDK4		
212	CDKN2A	NM_000077.4	trans-var-1
213	CDKN2A	NM_058197.4	trans-var-3
214	CDKN2A	NM_058195.3	trans-var-4
215	CDKN2A	NM_001195132.1	trans-var-5
216	CDKN2A		
217	CDKN2A	Exon 1	501-956
218	CDKN2A	Exon 2	501-807
219	CDKN2A	Exon 3	501-697
220	CDKN2A	Exon 4	501-991
221	CDKN2A		
222	CEBPA	NM_004364.3	
223	CHEK2	NM_007194.3	trans-var-1
224	CHEK2	NM_145862.2	trans-var-2
225	CHEK2	NM_001005735.1	trans-var-3
226	CHEK2		
227	CHEK2	Exon 1	501-566
228	CHEK2	Exon 2	501-825
229	CHEK2	Exon 3	501-629
230	CHEK2	Exon 4	501-625

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
231	CHEK2	Exon 5	501-648
232	CHEK2	Exon 6	501-591
233	CHEK2	Exon 7	501-609
234	CHEK2	Exon 8	501-554
235	CHEK2	Exon 9	501-562
236	CHEK2	Exon 10	501-600
237	CHEK2	Exon 11	501-587
238	CHEK2	Exon 12	501-664
239	CHEK2	Exon 13	501-616
240	CHEK2	Exon 14	501-586
241	CHEK2	Exon 15	501-581
242	CHEK2	Exon 16	501-744
243	CHEK2		
244	EPCAM	NM_002354.2	
245	EPCAM		
246	EPCAM	Exon 1	501-934
247	EPCAM	Exon 2	501-608
248	EPCAM	Exon 3	501-741
249	EPCAM	Exon 4	501-566
250	EPCAM	Exon 5	501-564
251	EPCAM	Exon 6	501-602
252	EPCAM	Exon 7	501-701
253	EPCAM	Exon 8	501-545
254	EPCAM	Exon 9	501-957
255	EPCAM		
256	FANCA	NM_000135.2	trans-var-1
257	FANCA	NM_001018112.1	trans-var-2
258	FANCB	NM_001018113.1	trans-var-1
259	FANCB	NM_152633.2	trans-var-2
260	FANCC	NM_000136.2	trans-var-1
261	FANCC	NM_001243743.1	trans-var-2
262	FANCC	NM_001243744.1	trans-var-3
263	FANCD2	NM_033084.3	trans-var-1
264	FANCD2	NM_001018115.1	trans-var-2
265	FANCE	NM_021922.2	
266	FANCF	NM_022725.3	
267	FANCG	NM_004629.1	
268	FANCI	NM_001113378.1	trans-var-1
269	FANCI	NM_018193.2	trans-var-2
270	FANCL	NM_001114636.1	trans-var-1
271	FANCL	NM_018062.3	trans-var-2
272	FANCM	NM_020937.2	
273	FLCN	NM_144997.5	trans-var-1
274	FLCN	NM_144606.5	trans-var-2
275	HOXB13	NM_006361.5	
276	HOXB13		
277	HOXB13	Exon 1	501-1257
278	HOXB13	Exon 2	501-2779
279	HOXB13		
280	KRAS	NM_033360.2	trans-var-a
281	KRAS	NM_004985.3	trans-var-b
282	MEN1	NM_000244.3	trans-var-1
283	MEN1	NM_130799.2	trans-var-2
284	MEN1	NM_130800.2	trans-var-e1B
285	MEN1	NM_130801.2	trans-var-e1C
286	MEN1	NM_130802.2	trans-var-e1D
287	MEN1	NM_130803.2	trans-var-e1E
288	MEN1	NM_130804.2	trans-var-e1F1
289	MLH1	NM_000249.3	trans-var-1
290	MLH1	NM_001167617.1	trans-var-2
291	MLH1	NM_001167618.1	trans-var-3
292	MLH1	NM_001167619.1	trans-var-4
293	MLH1		
294	MLH1	Exon 1	501-814
295	MLH1	Exon 2	501-591
296	MLH1	Exon 3	501-599
297	MLH1	Exon 4	501-574
298	MLH1	Exon 5	501-573
299	MLH1	Exon 6	501-592
300	MLH1	Exon 7	501-543
301	MLH1	Exon 8	501-589
302	MLH1	Exon 9	501-613
303	MLH1	Exon 10	501-594
304	MLH1	Exon 11	501-654

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
305	MLH1	Exon 12	501-871
306	MLH1	Exon 13	501-649
307	MLH1	Exon 14	501-609
308	MLH1	Exon 15	501-564
309	MLH1	Exon 16	501-665
310	MLH1	Exon 17	501-593
311	MLH1	Exon 18	501-614
312	MLH1	Exon 19	501-861
313	MLH1		
314	MLH3	NM_001040108.1	trans-var-1
315	MLH3	NM_014381.2	trans-var-2
316	MLH3		
317	MLH3	Exon 1	501-653
318	MLH3	Exon 2	501-3843
319	MLH3	Exon 3	501-599
320	MLH3	Exon 4	501-586
321	MLH3	Exon 5	501-605
322	MLH3	Exon 6	501-573
323	MLH3	Exon 7	501-572
324	MLH3	Exon 8	501-612
325	MLH3	Exon 9	501-660
326	MLH3	Exon 10	501-524
327	MLH3	Exon 11	501-579
328	MLH3	Exon 12	501-652
329	MLH3	Exon 13	501-3938
330	MLH3		
331	MRE11A	NM_005591.3	trans-var-1
332	MRE11A	NM_005590.3	trans-var-2
333	MRE11A		
334	MRE11A	Exon 1	501-584
335	MRE11A	Exon 2	501-625
336	MRE11A	Exon 3	501-633
337	MRE11A	Exon 4	501-661
338	MRE11A	Exon 5	501-588
339	MRE11A	Exon 6	501-642
340	MRE11A	Exon 7	501-615
341	MRE11A	Exon 8	501-686
342	MRE11A	Exon 9	501-672
343	MRE11A	Exon 10	501-581
344	MRE11A	Exon 11	501-627
345	MRE11A	Exon 12	501-601
346	MRE11A	Exon 13	501-674
347	MRE11A	Exon 14	501-563
348	MRE11A	Exon 15	501-720
349	MRE11A	Exon 16	501-584
350	MRE11A	Exon 17	501-559
351	MRE11A	Exon 18	501-568
352	MRE11A	Exon 19	501-576
353	MRE11A	Exon 20	501-3379
354	MRE11A		
355	MSH2	NM_000251.1	
356	MSH2		
357	MSH2	Exon 1	501-779
358	MSH2	Exon 2	501-655
359	MSH2	Exon 3	501-779
360	MSH2	Exon 4	501-647
361	MSH2	Exon 5	501-650
362	MSH2	Exon 6	501-634
363	MSH2	Exon 7	501-700
364	MSH2	Exon 8	501-610
365	MSH2	Exon 9	501-624
366	MSH2	Exon 10	501-651
367	MSH2	Exon 11	501-598
368	MSH2	Exon 12	501-746
369	MSH2	Exon 13	501-705
370	MSH2	Exon 14	501-748
371	MSH2	Exon 15	501-676
372	MSH2	Exon 16	501-943
373	MSH2		
374	MSH6	NM_000179.2	
375	MSH6		
376	MSH6	Exon 1	501-912
377	MSH6	Exon 2	501-697
378	MSH6	Exon 3	501-670

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID	5
379	MSH6	Exon 4	501-3045	10
380	MSH6	Exon 5	501-766	
381	MSH6	Exon 6	501-618	
382	MSH6	Exon 7	501-590	
383	MSH6	Exon 8	501-654	
384	MSH6	Exon 9	501-700	
385	MSH6	Exon 10	501-675	15
386	MSH6			
387	MUTYH	NM_012222.2	trans-var-alpha1	
388	MUTYH	NM_001048171.1	trans-var-alpha3	
389	MUTYH	NM_001128425.1	trans-var-alpha5	
390	MUTYH	NM_001048174.1	trans-var-beta3	
391	MUTYH	NM_001048172.1	trans-var-gamma2	20
392	MUTYH	NM_001048173.1	trans-var-gamma3	
393	MUTHY			
394	MUTHY	Exon 1	501-752	
395	MUTHY	Exon 2	501-621	
396	MUTHY	Exon 3	501-691	
397	MUTHY	Exon 4	501-540	25
398	MUTHY	Exon 5	501-574	
399	MUTHY	Exon 6	501-542	
400	MUTHY	Exon 7	501-572	
401	MUTHY	Exon 8	501-614	
402	MUTHY	Exon 9	501-598	
403	MUTHY	Exon 10	501-645	30
404	MUTHY	Exon 11	501-564	
405	MUTHY	Exon 12	501-689	
406	MUTHY	Exon 13	501-637	
407	MUTHY	Exon 14	501-653	
408	MUTHY	Exon 15	501-542	
409	MUTHY	Exon 16	501-696	35
410	MUTHY			
411	NBN	NM_002485.4		
412	NBN			
413	NBN	Exon 1	501-647	
414	NBN	Exon 2	501-634	
415	NBN	Exon 3	501-649	40
416	NBN	Exon 4	501-660	
417	NBN	Exon 5	501-604	
418	NBN	Exon 6	501-618	
419	NBN	Exon 7	501-694	
420	NBN	Exon 8	501-598	
421	NBN	Exon 9	501-630	45
422	NBN	Exon 10	501-773	
423	NBN	Exon 11	501-948	
424	NBN	Exon 12	501-569	
425	NBN	Exon 13	501-656	
426	NBN	Exon 14	501-614	
427	NBN	Exon 15	501-550	50
428	NBN	Exon 16	501-2777	
429	NBN			
430	PALB2	NM_024675.3		
431	PALB2			
432	PALB2	Exon 1	501-748	
433	PALB2	Exon 2	501-560	55
434	PALB2	Exon 3	501-603	
435	PALB2	Exon 4	501-1973	
436	PALB2	Exon 5	501-1330	
437	PALB2	Exon 6	501-572	
438	PALB2	Exon 7	501-662	
439	PALB2	Exon 8	501-586	60
440	PALB2	Exon 9	501-662	
441	PALB2	Exon 10	501-617	
442	PALB2	Exon 11	501-588	
443	PALB2	Exon 12	501-649	
444	PALB2	Exon 13	501-1008	
445	PALB2			65
446	PIK3CA	NM_006218.2		
447	PMS1	NM_000534.4	trans-var-1	
448	PMS1	NM_001128143.1	trans-var-2	
449	PMS1	NM_001128144.1	trans-var-3	
450	PMS2	NM_000535.5		
451	PMS2			65
452	PMS2	Exon 1	501-610	

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID	5
453	PMS2	Exon 2	501-640	10
454	PMS2	Exon 3	501-587	
455	PMS2	Exon 4	501-603	
456	PMS2	Exon 5	501-684	
457	PMS2	Exon 6	501-668	
458	PMS2	Exon 7	501-598	
459	PMS2	Exon 8	501-600	15
460	PMS2	Exon 9	501-585	
461	PMS2	Exon 10	501-656	
462	PMS2	Exon 11	501-1362	
463	PMS2	Exon 12	501-668	
464	PMS2	Exon 13	501-601	20
465	PMS2	Exon 14	501-670	
466	PMS2	Exon 15	501-804	
467	PMS2			
468	PTCH1	NM_001083602.1	trans-var-1a	
469	PTCH1	NM_001083603.1	trans-var-1a'	25
470	PTCH1	NM_000264.3	trans-var-1b	
471	PTCH1	NM_001083604.1	trans-var-1c	
472	PTCH1	NM_001083605.1	trans-var-1c'	
473	PTCH1	NM_001083606.1	trans-var-1d	
474	PTCH1	NM_001083607.1	trans-var-1e	
475	PTEN	NM_000314.4		30
476	PTEN			
477	PTEN	Exon 1	501-1611	
478	PTEN	Exon 2	501-585	
479	PTEN	Exon 3	501-545	
480	PTEN	Exon 4	501-544	
481	PTEN	Exon 5	501-739	35
482	PTEN	Exon 6	501-642	
483	PTEN	Exon 7	501-667	
484	PTEN	Exon 8	501-725	
485	PTEN	Exon 9	501-3989	
486	PTEN			
487	RAD50	NM_005732.3		40
488	RAD50			
489	RAD50	Exon 1	501-1030	
490	RAD50	Exon 2	501-584	
491	RAD50	Exon 3	501-651	
492	RAD50	Exon 4	501-686	
493	RAD50	Exon 5	501-705	45
494	RAD50	Exon 6	501-629	
495	RAD50	Exon 7	501-666	
496	RAD50	Exon 8	501-694	
497	RAD50	Exon 9	501-707	
498	RAD50	Exon 10	501-683	
499	RAD50	Exon 11	501-685	50
500	RAD50	Exon 12	501-676	
501	RAD50	Exon 13	501-738	
502	RAD50	Exon 14	501-690	
503	RAD50	Exon 15	501-627	
504	RAD50	Exon 16	501-694	
505	RAD50	Exon 17	501-611	55
506	RAD50	Exon 18	501-593	
507	RAD50	Exon 19	501-614	
508	RAD50	Exon 20	501-628	
509	RAD50	Exon 21	501-725	
510	RAD50	Exon 22	501-586	
511	RAD50	Exon 23	501-643	60
512	RAD50	Exon 24	501-634	
513	RAD50	Exon 25	501-2944	
514	RAD50			
515	RAD51C	NM_058216.1	trans-var-1	
516	RAD51C	NM_002876.2	trans-var-2	65
517	RAD51C			
518	RAD51C	Exon 1	501-687	
519	RAD51C	Exon 2	501-904	
520	RAD51C			
521	RAD51D	NM_002878.3	trans-var-1	
522	RAD51D	NM_133629.2	trans-var-4	65
523	RAD51D	NM_001142571.1	trans-var-6	
524	RAD51D			
525	RAD51D	Exon 1	501-838	
526	RAD51D	Exon 2	501-562	

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
527	RAD51D	Exon 3	501-679
528	RAD51D	Exon 4	501-619
529	RAD51D	Exon 5	501-582
530	RAD51D	Exon 6	501-635
531	RAD51D	Exon 7	501-596
532	RAD51D	Exon 8	501-591
533	RAD51D	Exon 9	501-571
534	RAD51D	Exon 10	501-665
535	RAD51D	Exon 11	501-1745
536	RAD51D		
537	RET	NM_020975.4	trans-var-2
538	RET	NM_020630.4	trans-var-4
539	SDHAF2	NM_017841.2	
540	SDHB	NM_003000.2	
541	SDHC	NM_003001.3	trans-var-1
542	SDHC	NM_001035511.1	trans-var-2
543	SDHC	NM_001035512.1	trans-var-3
544	SDHC	NM_001035513.1	trans-var-4
545	SDHD	NM_003002.2	
546	SMAD4	NM_005359.5	
547	SMAD4		
548	SMAD4	Exon 1	501-911
549	SMAD4	Exon 2	501-876
550	SMAD4	Exon 3	501-675
551	SMAD4	Exon 4	501-530
552	SMAD4	Exon 5	501-713
553	SMAD4	Exon 6	501-620
554	SMAD4	Exon 7	501-617
555	SMAD4	Exon 8	501-551
556	SMAD4	Exon 9	501-684
557	SMAD4	Exon 10	501-669
558	SMAD4	Exon 11	501-639
559	SMAD4	Exon 12	501-7286
560	SMAD4		
561	STK11	NM_000455.4	
562	STK11		
563	STK11	Exon 1	501-1905
564	STK11	Exon 2	501-584
565	STK11	Exon 3	501-590
566	STK11	Exon 4	501-633
567	STK11	Exon 5	501-637
568	STK11	Exon 6	501-628
569	STK11	Exon 7	501-558
570	STK11	Exon 8	501-688
571	STK11	Exon 9	501-710
572	STK11	Exon 10	501-1343
573	STK11		
574	TMEM127	NM_017849.3	trans-var-1
575	TMEM127	NM_001193304.2	trans-var-2
576	TP53	NM_000546.4	trans-var-1
577	TP53	NM_001126112.2	trans-var-2
578	TP53	NM_001126114.1	trans-var-3
579	TP53	NM_001126113.1	trans-var-4
580	TP53	NM_001126115.1	trans-var-5
581	TP53	NM_001126116.1	trans-var-6
582	TP53	NM_001126117.1	trans-var-7
583	TP53		
584	TP53		
585	TP53	Exon 1	501-674
586	TP53	Exon 2	501-602
587	TP53	Exon 3	501-741
588	TP53	Exon 4	501-522
589	TP53	Exon 5	501-779
590	TP53	Exon 6	501-684
591	TP53	Exon 7	501-613
592	TP53	Exon 8	501-941
593	TP53	Exon 9	501-610
594	TP53	Exon 10	501-637
595	TP53	Exon 11	501-574
596	TP53	Exon 12	501-607
597	TP53	Exon 13	501-560
598	TP53	Exon 14	501-633
599	TP53	Exon 15	501-1789
600	TP53		

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TABLE 3-continued

SEQ ID NO	Entrez Gene Symbol	RefSeq Accession #	Transcript Variant or Exon Coordinates in SEQ ID
601	VHL	NM_000551.3	trans-var-1
602	VHL	NM_198156.2	trans-var-2

Table 3 shows how sequence identifiers (i.e., SEQ ID NOs) correspond to different reference sequences useful for the various HCGs in various aspects of the invention. As used in Table 3, “transcript variant” refers to differently spliced transcripts expressed from some genes. In cases where no transcript variant is indicated, this is because NCBI lists only one transcript for the relevant gene. The exon coordinates given in Table 3 indicate where in each relevant sequence the exons are found. The first 500 and last 500 nucleotides of each such sequence are intronic. As used herein, “exon/intron boundary” in one of these sequences means a certain number of nucleotides (e.g., 1, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50, 75, 100 or more) on each side of the transition (e.g., phosphodiester bond) from exon to intron (or from intron to exon) or a portion of the nucleotide sequence of at least a certain length (e.g., 1, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50, 75, 100 or more) comprising the two nucleotides on each side of the transition from exon to intron (or from intron to exon).

In some embodiments of various aspects of the invention, a nucleic acid of the invention {e.g., in a primer set, in an array, in a kit, etc.) comprises at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, or 100 or more nucleotides on each side of such transition. Thus, an oligonucleotide {e.g., primer) according to the invention targeting Exon 3 of the APC gene “comprising 10 nucleotides on each side of the 5' exon/intron boundary of Exon 3 of the APC gene” would comprise nucleotides 491-510 of SEQ ID NO:7, or the following sequence: 5'-ttttatttagAGCTTAAGT-3' (SEQ ID NO: 610; with lower case letters indicating intronic sequence and capitalized letters indicating exonic sequence). In some embodiments of various aspects of the invention, a nucleic acid of the invention comprises at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, or 100 or more consecutive nucleotides of a nucleotide sequence in a SEQ ID NO including the two nucleotides on each side of such transition. Thus, an oligonucleotide (e.g., primer) according to the invention targeting Exon 3 of the APC gene “comprising 18 consecutive nucleotides of SEQ ID NO: 7 including the 5' exon/intron boundary of Exon 3 of the APC gene” would comprise any 18 consecutive nucleotides between (and including) positions 484 and 517 of SEQ ID NO:7, or any 5'-18 consecutive nucleotides of the following sequence: gttctattttatttagAGCTTAAGTATAGC-3' (SEQ ID NO: 611; with lower case letters indicating intronic sequence and capitalized letters indicating exonic sequence). At various places in this document Exon 3 of the APC gene is used as an example to illustrate various embodiments of the invention. Those skilled in the art, based on the knowledge in the art and the present disclosure (especially Table 3), can readily and unambiguously apply each example to any gene, exon, or sequence disclosed herein.

Germline deficiencies in the genes in Panels A-Q correlate to increased risk of cancer, including particular cancers as summarized in Table 4. Thus, in some embodiments the

method of the invention comprises correlating a germline deficiency in any particular gene in the plurality of genes to an increased risk of a particular cancer as shown in Table 4. In some embodiments the method comprises diagnosing the patient with an increased risk of a particular cancer (or a particular syndrome) as shown in Table 4 based at least in part on a germline deficiency in any particular gene in the plurality of genes. In some embodiments the method comprises correlating no germline deficiency in any gene in the plurality of genes to no increased risk of any cancer (or to no identified increased risk due to the tested genes). In some embodiments the system of the invention comprises a computer program for determining (including quantifying) the patient's degree of risk of cancer (e.g., any particular cancer as shown in Table 4) based at least in part on the comparison of the test sequence with said one or more reference sequences.

TABLE 4

Gene Symbol	Associated Cancer (e.g., indicator of syndrome or hereditary cancer risk)	Syndrome (if any)
APC	Colon	FAP
ATM	Breast	Ataxia Telangiectasia
BARD1	Breast	
BMPR1A	GI	Juvenile Polyposis Syndrome
BRCA1	Breast, Ovarian	Hereditary Breast and Ovarian Cancer Syndrome (HBOC)
BRCA2	Breast, Ovarian	HBOC
BRIP1	Breast,	
CDH1	Breast, Gastric	Hereditary Diffuse Gastric Cancer
CDK4	Melanoma	Hereditary Melanoma (aka Multiple Nevi Syndrome)
CDKN2A	Melanoma, Pancreatic	Hereditary Melanoma (aka Multiple Nevi Syndrome)
CHEK2	Breast, Colon	
HOXB13	Prostate	
MLH1	Colon, Endometrial, Ovarian	Lynch Syndrome (aka Hereditary Non-Polyposis Colorectal Cancer or HNPCC)
MLH3	Colon, Endometrial, Ovarian	Lynch Syndrome
MRE11		
MSH2	Colon, Endometrial, Ovarian	Lynch Syndrome
MSH6	Colon, Endometrial, Ovarian	Lynch Syndrome
MUTYH	Colon	MYH-associated polyposis
NBN	Breast	
PALB2	Pancreatic, Breast	
PMS2	Colon, Endometrial, Ovarian	Lynch Syndrome
PTEN	Breast, Endometrial	Cowden Syndrome
RAD50	Breast	
RAD51C	Breast, Ovarian	HBOC
RAD51D	Ovarian	HBOC
SMAD4	GI	Juvenile Polyposis Syndrome
STK11	GI, Breast	Peutz-Jeghers Syndrome
EPCAM	Colon, Endometrial, Ovarian	Lynch Syndrome
TP53	Breast, Brain, Sarcoma	Li-Fraumeni Syndrome

In some embodiments the panel of the invention to be assessed in a particular patient depends on the specific cancer(s) or syndrome(s) for which the patient is apparently at risk. For example, as shown in Example 2 below, a patient presenting with indicators of HBOC may be tested for a panel of test genes comprising Panel D (or Panel P) or any subpanel comprising the top 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 genes of Panel D (or Panel P). Thus, in some embodiments of the methods and systems described above the patient is identified as having one or more indicators of a syndrome listed in Table 4, or otherwise having one or more indicators of an increased predisposition to one or more of the cancers listed in Table 4, and the patient is tested for a

panel comprising genes whose mutations are associated with that syndrome or cancer. In some embodiments an indicator of a particular syndrome listed in Table 4 is present when the patient has one or more of the corresponding cancers listed in Table 4 (e.g., an indicator of Lynch syndrome may be endometrial cancer in the patient).

In some embodiments the genes of Panel P may be added iteratively to BRCA1 and BRCA2, which may include reflex testing later genes upon determining the patient is negative for earlier genes. In some embodiments the panel of test genes comprises BRCA1, BRCA2 and CHEK2. In some embodiments, the panel of test genes comprises BRCA1, BRCA2, CHEK2; and any one, two or three of ATM, NBN and/or PALB2. In some embodiments, the panel of test genes comprises BRCA1, BRCA2, CHEK2; any one, two or three of ATM, NBN and/or PALB2; and any one or two of BARD1 and/or BRIP1. In some embodiments, the panel of test genes comprises BRCA1, BRCA2, CHEK2; any one, two or three of ATM, NBN and/or PALB2; any one or two of BARD1 and/or BRIP1; and PMS2. In some embodiments, the panel of test genes comprises BRCA1, BRCA2, CHEK2; any one, two or three of ATM, NBN and/or PALB2; any one or two of BARD1 and/or BRIP1; PMS2; any one, two or three of MSH2, MSH6 and/or TP53. In some embodiments, the panel of test genes comprises BRCA1, BRCA2, CHEK2; any one, two or three of ATM, NBN and/or PALB2; any one or two of BARD1 and/or BRIP1; PMS2; any one, two or three of MSH2, MSH6 and/or TP53; and MUTYH.

In some embodiments, the invention provides a method of diagnosing increased risk of breast or ovarian cancer comprising (1) identifying the patient as having at least one indicator of a genetic predisposition to breast or ovarian cancer; (2) assaying a sample from the patient to detect one or more mutations in a plurality of test genes comprising at least 3 (e.g., the top 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12) genes in Panel D; and (3)(a) diagnosing the patient as having an increased risk of breast or ovarian cancer if a mutation is detected in step (2) or (3)(b) diagnosing the patient as not having an increased risk of breast or ovarian cancer if no mutation is detected in step (2).

In some embodiments, an indicator of genetic predisposition to breast or ovarian cancer is any of the following:

Personal and/or family history of ovarian cancer;
Personal and/or family history of breast cancer (e.g., diagnosed before a certain age (e.g., 35, 40, 45, 50, 55, 60, 65 or 70));

Personal and/or family history of two primary breast cancers;

Personal and/or family history of male breast cancer;
Personal and/or family history of triple negative breast cancer;

Ashkenazi Jewish descent with personal and/or family history of breast, ovarian, pancreatic, or aggressive prostate cancer (Gleason score of >7);

Personal and/or family history of three or more cancers chosen from breast, ovarian, pancreatic, or aggressive prostate cancer (Gleason score of >7); or

A previously identified mutation in any close blood relative in any of the at least 3 genes from Panel D.

As used above, "breast cancer" includes both invasive cancer and ductal carcinoma in situ (DCIS) and "ovarian cancer" includes epithelial ovarian cancer, fallopian tube cancer, and primary peritoneal cancer. As used above, "personal history" of any of these indicators means patient has been identified as having the indicator (e.g., the patient has been diagnosed as having triple negative breast cancer). As

used above, “family history” of any of these indicators means a close blood relative having such indicator and “close blood relative” means a 1st, 2nd, or 3rd degree relative in either the maternal or paternal lineage.

In some embodiments, the invention provides a method of diagnosing increased risk of a Lynch syndrome cancer comprising (1) identifying the patient as having at least one indicator of a genetic predisposition to a Lynch syndrome cancer; (2) assaying a sample from the patient to detect one or more mutations in a plurality of test genes comprising at least 3 (e.g., the top 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12) genes in Panel E; and (3)(a) diagnosing the patient as having an increased risk of the cancer if a mutation is detected in step (2) or (3)(b) diagnosing the patient as not having an increased risk of the cancer if no mutation is detected in step (2). As described in Example 3 below, the inventors have made the surprising discovery that mutations in BRCA1 and BRCA2 make a significant contribution to patients having Lynch syndrome. Thus in some embodiments the plurality of test genes comprises (a) MLH1, BRCA1, BRCA2; (b) MLH1, MSH2, BRCA1, BRCA2; (c) MLH1, MSH2, MSH6, BRCA1, BRCA2; (d) MLH1, MSH2, PMS2, BRCA1, BRCA2; (e) MLH1, MSH2, MUTYH, BRCA1, BRCA2; (f) MLH1, MSH2, MSH6, PMS2, BRCA1, BRCA2; (g) MLH1, MSH2, MSH6, PMS2, MUTYH, BRCA1, BRCA2; or (g) MLH1, MSH2, MSH6, PMS2, MUTYH, EPCAM, BRCA1, BRCA2.

In some embodiments, an indicator of genetic predisposition to a Lynch syndrome cancer is any of the following:

Personal and/or family history of colorectal or endometrial cancer (e.g., before a certain age (e.g., 35, 40, 45, 50, 55, 60, 65 or 70));

Personal and/or family history of colorectal cancer with MSI High histology (e.g., before a certain age (e.g., 35, 40, 45, 50, 55, 60, 65 or 70)), with examples of MSI high histology including any of the following:

Mucinous

Signet ring

Tumor infiltrating lymphocytes

Crohn's-like lymphocytic reaction

Medullary growth pattern;

Personal and/or family history of colorectal or endometrial cancer with abnormal MSIHC tumor test result;

Personal and/or family history of two or more Lynch syndrome cancers, including cases where at least one is before a certain age (e.g., 35, 40, 45, 50, 55, 60, 65 or 70);

Personal history of Lynch syndrome cancer with family history of a Lynch syndrome cancer;

Three or more close blood relatives with a Lynch syndrome cancer; or

A previously identified mutation in any close blood relative in any of the at least 3 genes from Panel E.

As used above, “Lynch syndrome cancer” may include any of the following: colorectal cancer, endometrial cancer, gastric cancer, ovarian cancer, ureter/renal pelvic cancer, biliary tract cancer, small bowel cancer, pancreatic cancer, brain cancer, or sebaceous adenomas. As used above, “personal history” of any of these indicators means patient has been identified as having the indicator (e.g., the patient has been diagnosed as having endometrial cancer). As used above, “family history” of any of these indicators means a close blood relative having such indicator and “close blood relative” means a 1st, 2nd, or 3rd degree relative in either the maternal or paternal lineage.

The nucleic acids to be analyzed in the methods and systems of the invention may vary in size. Thus, in some

embodiments A=10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, or 90,000, or more and B=15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, 90,000, or 100,000 or more. These embodiments include every combination of A and B as set forth in the preceding sentence, where B>A. For example, the nucleic acids to be analyzed may comprise (or consist of or consist essentially of) a range of nucleotides in length from any A to any B (e.g., from 10 to 15, 10 to 20, [. . .] 100 to 125, 100 to 150, etc.).

In some embodiments the plurality of DNA molecules comprises at least some length of intronic sequence adjacent to some (or all) of said one or more exons. In some embodiments, the plurality of DNA molecules comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 200, 300, 400, 500 or more base pairs of the intronic sequence on one or both sides of the exon(s). This may comprise some portion of the sequences disclosed herein, using Table 3 as reference for where exons and introns begin and end. For example, in one embodiment the plurality of DNA molecules comprises the exons of, e.g., the APC gene plus at least 20 intronic nucleotides upstream and 10 intronic nucleotides downstream of each exon. For Exon 3 of APC, for example, this would mean the plurality of DNA molecules comprises Exon 3 (nucleotides 501-702 of SEQ ID NO:7) and further comprises the first 20 nucleotides of the intron upstream of Exon 3 (nucleotides 481-500 of SEQ ID NO:7) and the first 10 nucleotides of the intron downstream of Exon 3 (nucleotides 703-712 of SEQ ID NO:7). Those skilled in the art can apply this to the other genes, exons, and sequences referenced in Table 3.

As mentioned above, the nucleic acids to be analyzed in the methods and systems of the invention comprise one or more exons of a plurality of genes. As used herein, a plurality of nucleic acid molecules comprises a sequence or group of sequences if such plurality of molecules together comprises the sequence or group of sequences. Multiple molecules together comprise a single sequence when the non-redundant sequences of the multiple molecules comprise such sequence. For example, a plurality of molecules may comprise the sequence of Exon 3 of the APC gene, which is just over 200 nucleotides long, despite each molecule being no more than 60 nucleotides long. This is true if the non-redundant sequences from the plurality of molecules, when considered end to end, comprise the full sequence of Exon 3. This example is illustrated in FIG. 1, which shows how a plurality of DNA molecules can comprise Exon 3 of the APC gene plus 10 upstream and 10 downstream intronic nucleotides. No single molecule comprises all of Exon 3. When they are aligned, however, the non-redundant sequences of these molecules (underlined nucleotides in Read1 to Read6) “together” make up a sequence (Composite) that comprises Exon 3 of the APC gene plus 10 upstream and 10 downstream intronic nucleotides (underlined nucleotides of Composite). As illustrated in FIG. 1 (Read1 and Read2), the molecules to be analyzed may comprise additional moieties that may include addi-

tional nucleotides and nucleotide sequences, fluorescent labels, conjugated antibodies or other proteins. Such molecules may still “together” comprise sequence of interest if the non-redundant nucleotide sequences of the molecules end-to-end comprise that sequence.

The total number of genes analyzed in the methods, systems and kits of the invention may vary depending on resource and technical constraints. Thus, in some embodiments W=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 or more and X=3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, or 20,000 or more. These embodiments include every combination of W and X as set forth in the preceding sentence, where X>W. For example, the plurality of genes to be analyzed may comprise (or consist of or consist essentially of) a range of genes in number from any W to any X (e.g., from 10 to 15, 10 to 20, [. . .] 100 to 125, 100 to 150, etc.).

The plurality of genes analyzed in the methods, systems and kits of the invention will comprise at least some of the genes listed in Panels A-Q. Thus, in some embodiments the plurality of genes comprises at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 genes listed in Panels A-Q. In some embodiments the plurality of genes comprises gene numbers between Y and Z of any of Panels A-Q. In some such embodiments, Y=1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67 or 68 and Z=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69. In some embodiments, said plurality of genes comprises gene numbers 1 & 2, 2 & 3, 3 & 4, 4 & 5, 5 & 6, 6 & 7, 7 & 8, 8 & 9, 9 & 10, 10 & 11, 11 & 12, 12 & 13, 13 & 14, 14 & 15, 15 & 16, 16 & 17, 17 & 18, 18 & 19, 19 & 20, 20 & 21, 21 & 22, 22 & 23, 23 & 24, 24 & 25, 25 & 26, 26 & 27, 27 & 28, 28 & 29, 29 & 30, 30 & 31, 31 & 32, 32 & 33, 33 & 34, 34 & 35, 35 & 36, 36 & 37, 37 & 38, 38 & 39, 39 & 40, 40 & 41, 41 & 42, 42 & 43, 43 & 44, 44 & 45, 45 & 46, 46 & 47, 47 & 48, 48 & 49, 49 & 50, 50 & 51, 51 & 52, 52 & 53, 53 & 54, 54 & 55, 55 & 56, 56 & 57, 57 & 58, 58 & 59, 59 & 60, 60 & 61, 61 & 62, 62 & 63, 63 & 64, 64 & 65, 65 & 66, 66 & 67, 67 & 68, or 68 & 69 from any of Panels A-Q. These embodiments include every combination of Y and Z as set forth in the preceding sentences, where Y>Z. For example, the plurality of genes to be analyzed may comprise (or consist of or consist essentially of) a range of genes with a number from any Y to any Z in any of Panels A-Q (e.g., from 1 to 2, 1 to 3, 1 to 4, [. . .] 1 to 55, 2 to 3, 2 to 4, 2 to 5, [. . .] 2 to 55, etc.). In some embodiments the genes chosen from Panels A-Q comprise at least some percentage, e.g., 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%,

90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%, of the plurality of genes to be analyzed.

In some embodiments the plurality of DNA molecules comprises at least some length of intronic sequence adjacent to some (or all) of said one or more exons. In some embodiments, the plurality of DNA molecules comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 200, 300, 400, 500 or more base pairs of the intronic sequence.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of BRCA1, BRCA2, PTEN, PALB2, CHEK2, BRIP1, BARD1, CDH1, ATM, RAD50, MRE11A, NBN, RAD51C, TP53, or STK11. In some embodiments, the plurality of genes comprises BRCA1, BRCA2, PTEN, PALB2, CHEK2, BRIP1, BARD1, CDH1, ATM, RAD50, MRE11A, NBN, RAD51C, TP53, and STK11 together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of MLH1, MSH2, MSH6, PMS2, EPCAM, APC or MUTYH. In some embodiments, the plurality of genes comprises MLH1, MSH2, MSH6, PMS2, EPCAM, APC and MUTYH together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of BRCA1, BRCA2, BRIP1, BARD1, CHEK2, MRE11A, NBN, RAD50, RAD51C, PALB2, TP53, PTEN, STK11, CDH1, ATM, MLH1, MSH2, MSH6, PMS1, PMS2 or MUTYH. In some embodiments, the plurality of genes comprises BRCA1, BRCA2, BRIP1, BARD1, CHEK2, MRE11A, NBN, RAD50, RAD51C, PALB2, TP53, PTEN, STK11, CDH1, ATM, MLH1, MSH2, MSH6, PMS1, PMS2 and MUTYH together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of PTEN, PALB2, STK11, CHEK2, ATM or TP53. In some embodiments, the plurality of genes comprises PTEN, PALB2, STK11, CHEK2, ATM and TP53 together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of MLH1, MSH2, MSH6, PMS2 or EPCAM. In some embodiments, the plurality of genes comprises MLH1, MSH2, MSH6, PMS2 and EPCAM together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of MLH1, MSH2, MSH6, or PMS2. In some embodiments, the plurality of genes comprises MLH1, MSH2, MSH6, and PMS2 together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of ACCA, COMT, CYP11B2, CYP 19, CYP1A1, CYP1B1, EPHX, ERA, FASL, IGF2, INS, KLK10, MSH6, RAD51L3, SOD2, VDR, XPG, or XRCC2. In some embodiments, the plurality of genes comprises ACCA, COMT, CYP11B2, CYP 19, CYP1A1, CYP1B1, EPHX, ERA, FASL, IGF2, INS, KLK10, MSH6, RAD51L3, SOD2, VDR, XPG, and XRCC2 together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of BRCA1, BRCA2, CHEK2, RAD51, or NBN. In some embodiments, the plurality of genes comprises BRCA1, BRCA2, CHEK2, RAD51, and NBN together with at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more additional gene(s) (including gene number(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10) from any of Panels A-Q.

In some embodiments, the plurality of genes comprises the genes in any of Panels A-Q, with the proviso that the genes do not include one or more of ABL1, AKT1, ALK, APC, ATM, BRAF, CDH1, CDKN2A, CSF1R, CTNNB1, EGFR, ERBB2, ERBB4, FBXW7, FGFR1, FGFR2, FGFR3, FLT3, GNAS, HNF1A, HRAS, IDH1, JAK2, JAK3, KDR, KIT, KRAS, MET, MLH1, MPL, NOTCH1, NPM1, NRAS, PDGFRA, PIK3CA, PTEN, PTPN11, RB1, RET, SMAD4, SMARCB1, SMO, SRC, STK11, TP53, or VHL. In some embodiments, the plurality of genes comprises ABL1, AKT1, ALK, APC, ATM, BRAF, CDH1, CDKN2A, CSF1R, CTNNB1, EGFR, ERBB2, ERBB4, FBXW7, FGFR1, FGFR2, FGFR3, FLT3, GNAS, HNF1A, HRAS, IDH1, JAK2, JAK3, KDR, KIT, KRAS, MET, MLH1, MPL, NOTCH1, NPM1, NRAS, PDGFRA, PIK3CA, PTEN, PTPN11, RB1, RET, SMAD4, SMARCB1, SMO, SRC, STK11, TP53, and VHL together with at least one additional gene from any of Panels A-Q.

As used herein, a “deficiency” in a gene means the presence of some sequence, copy number, expression or epigenetic variation from wild-type in the gene that leads to a deleterious change in function. Sequence variations include point mutations, small (e.g., less than 1,000 nucleotides) deletions and insertions (including frameshift mutations), large (e.g., greater than 1,000 nucleotides) deletions and insertions, and transversions (e.g., reversal of direction in a region of the gene). Copy number variations include amplifications and deletions of substantially an entire gene. Epigenetic variations include variations in methylation, acetylation, etc. In the case of tumor suppressors, a deleterious change in function will generally be attenuated function. Examples include lowered or abolished transcription, lowered or abolished protein expression, and lowered or abolished protein function. Many variations that will lead to such changes may be recognized by those skilled in the art based on the present disclosure, including frameshift or nonsense (premature stop) mutations; deletions, amplifications or transversions in large regions of the gene; missense mutations in critical interaction, structural or enzymatic regions; etc. In the case of oncogenes, a deleterious change in function will generally be heightened function. Examples include heightened transcription, heightened protein expression, and heightened protein function. Many variations that will lead to such changes may be recognized by those skilled

in the art based on the present disclosure, including amplification of the gene and activating mutations in enzymatic regions.

As used herein, a “germline” deficiency is any deficiency that is found in the germline of the individual as opposed to deficiencies found only in somatic tissues. For example, a deficiency found in a tumor tissue may either have originated in the germline or arisen somatically. Germline deficiencies may be detected by analyzing various types of samples. Generally, these samples will contain or be derived from cells expected to represent the germline. Examples include white blood cells, germ cells, etc. In some embodiments the nucleic acid analyzed is genomic DNA from such a cell (or DNA (e.g., PCR amplified DNA) derived therefrom). In other embodiments, the nucleic acid analyzed is transcript RNA (or complementary DNA transcribed therefrom) from such a cell. In some embodiments, protein derived from such a cell is analyzed for structural (e.g., amino acid sequence) and functional deficiencies.

Those skilled in the art are familiar with various techniques for sequencing nucleic acids in a sample. Useful techniques include, but are not limited to, Sanger sequencing, sequencing by synthesis (e.g., as described in U.S. Pat. Nos. 6,828,100, 7,276,720, and 7,283,337 and U.S. application publication nos. US20110212437, US20110229877, US20110177498, US20120064599, and US20120058468), single-molecule sequencing (e.g., as described in U.S. Pat. Nos. 8,148,516 and 8,137,569 and U.S. application publication nos. US20110212437, US20110229877, US20110177498, US20120064599, and US20120058468), etc. Examples include techniques developed by Applied Biosystems™ (SOLiD™), Illumina™ (HiSeq™), 454™, Pacific Biosciences™ (SMRT™), and Oxford Nanopore™ (GridION™ and MinION™), each of which is well-known to those skilled in the art.

As discussed above, the methods of the invention generally involve sequencing a panel of genes described herein. With modern techniques, it is often possible to sequence tens, hundreds or thousands of genes. Indeed, it is possible to sequence the entire genome. Once such a global assay has been performed, one may then informatically analyze one or more subsets of genes (i.e., panels or, as often used herein, pluralities of test genes). After sequencing hundreds or thousands of genes in a sample, for example, one may analyze (e.g., informatically) the sequences of a panel or plurality of test genes comprising primarily genes in any of Panels A-Q according to the present invention (e.g., to determine whether a patient has an increased risk of a particular cancer).

As used herein, a patient has an “increased risk” of a particular cancer if the probability of the patient developing that cancer (e.g., over the patient’s lifetime, over some defined period of time (e.g., within 10 years), etc.) exceeds some reference probability or value. The reference probability may be the probability (i.e., prevalence) of the cancer across the general relevant patient population (e.g., all patients; all patients of a particular age, gender, ethnicity; patients having a particular cancer (and thus looking at the risk of a different cancer or an independent second primary of the same type as the first cancer); etc.). For example, if the lifetime probability of a particular cancer in the general population (or some specific subpopulation) is X % and a particular patient has been determined by the methods, systems or kits of the present invention to have a lifetime probability of that cancer of Y %, and if Y>X, then the patient has an “increased risk” of that cancer. Alternatively, the tested patient’s probability may only be considered

“increased” when it exceeds the reference probability by some threshold amount (e.g., at least 0.5, 0.75, 0.85, 0.90, 0.95, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more fold or standard deviations greater than the reference probability; at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% greater than the reference probability).

The results of any analyses according to the invention will often be communicated to physicians, genetic counselors and/or patients (or other interested parties such as researchers) in a transmittable form that can be communicated or transmitted to any of the above parties. Such a form can vary and can be tangible or intangible. The results can be embodied in descriptive statements, diagrams, photographs, charts, images or any other visual forms. For example, graphs showing expression or activity level or sequence variation information for various genes can be used in explaining the results. Diagrams showing such information for additional target gene(s) are also useful in indicating some testing results. The statements and visual forms can be recorded on a tangible medium such as papers, computer readable media such as floppy disks, compact disks, etc., or on an intangible medium, e.g., an electronic medium in the form of email or website on internet or intranet. In addition, results can also be recorded in a sound form and transmitted through any suitable medium, e.g., analog or digital cable lines, fiber optic cables, etc., via telephone, facsimile, wireless mobile phone, internet phone and the like.

Thus, the information and data on a test result can be produced anywhere in the world and transmitted to a different location. As an illustrative example, when a sequencing (or genotyping) assay is conducted outside the United States, the information and data on a test result may be generated, cast in a transmittable form as described above, and then imported into the United States. Accordingly, the present invention also encompasses methods and systems for producing a transmittable form of sequence information for at least one patient sample. The method comprises the steps of (1) sequencing nucleic acids in a sample according to methods of the present invention; and (2) embodying the result of the sequencing step in a transmittable form. The transmittable form is a product of such a method.

Techniques for analyzing sequence data (indeed any data obtained according to the invention) will often be implemented using hardware, software or a combination thereof in one or more computer systems or other processing systems capable of effectuating such analysis.

The sample analyzer in the systems of the invention can be any instrument useful in sequencing nucleic acids, including but not limited to, Illumina HiSeg™, Ion Torrent PGM, ABI SOLiD™ sequencer, PacBio RS, Helicos Heliscope™, or any instrument utilizing a sequencing system discussed above.

The computer-based analysis function can be implemented in any suitable language and/or browsers. For example, it may be implemented with C language and preferably using object-oriented high-level programming languages such as Visual Basic, SmallTalk, C++, and the like. The application can be written to suit environments such as the Microsoft Windows™ environment including Windows™ 98, Windows™ 2000, Windows™ NT, and the like. In addition, the application can also be written for the Macintosh™, SUN™, UNIX or LINUX environment. In addition, the functional steps can also be implemented using a universal or platform-independent programming language.

Examples of such multi-platform programming languages include, but are not limited to, hypertext markup language (HTML), JAVA™, JavaScript™, Flash programming language, common gateway interface/structured query language (CGI/SQL), practical extraction report language (PERL), AppleScript™ and other system script languages, programming language/structured query language (PL/SQL), and the like. Java™- or JavaScript™-enabled browsers such as HotJava™, Microsoft™ Explorer™, or Netscape™ can be used. When active content web pages are used, they may include Java™ applets or ActiveX™ controls or other active content technologies.

The analysis function can also be embodied in computer program products and used in the systems described above or other computer- or internet-based systems. Accordingly, another aspect of the present invention relates to a computer program product comprising a computer-usable medium having computer-readable program codes or instructions embodied thereon for enabling a processor to carry out gene status analysis. These computer program instructions may be loaded onto a computer or other programmable apparatus to produce a machine, such that the instructions which execute on the computer or other programmable apparatus create means for implementing the functions or steps described above. These computer program instructions may also be stored in a computer-readable memory or medium that can direct a computer or other programmable apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory or medium produce an article of manufacture including instruction means which implement the analysis. The computer program instructions may also be loaded onto a computer or other programmable apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions or steps described above.

One example of a computer system of the invention is the computer system [200] illustrated in FIG. 2. Computer system [200] may include at least one input module [230] for entering patient data into the computer system [200]. The computer system [200] may include at least one output module [224] for indicating whether a patient has an increased or decreased likelihood of response and/or indicating suggested treatments determined by the computer system [200]. Computer system [200] may include at least one memory module [206] in communication with the at least one input module [230] and the at least one output module [224].

The at least one memory module [206] may include, e.g., a removable storage drive [208], which can be in various forms, including but not limited to, a magnetic tape drive, a floppy disk drive, a VCD drive, a DVD drive, an optical disk drive, etc. The removable storage drive [208] may be compatible with a removable storage unit [210] such that it can read from and/or write to the removable storage unit [210]. Removable storage unit [210] may include a computer usable storage medium having stored therein computer-readable program codes or instructions and/or computer readable data. For example, removable storage unit [210] may store patient data. Example of removable storage unit [210] are well known in the art, including, but not limited to, floppy disks, magnetic tapes, optical disks, and the like. The at least one memory module [206] may also

include a hard disk drive [212], which can be used to store computer readable program codes or instructions, and/or computer readable data.

In addition, as shown in FIG. 2, the at least one memory module [206] may further include an interface [214] and a removable storage unit [216] that is compatible with interface [214] such that software, computer readable codes or instructions can be transferred from the removable storage unit [216] into computer system [200]. Examples of interface [214] and removable storage unit [216] pairs include, e.g., removable memory chips (e.g., EPROMs or PROMs) and sockets associated therewith, program cartridges and cartridge interface, and the like. Computer system [200] may also include a secondary memory module [218], such as random access memory (RAM).

Computer system [200] may include at least one processor module [202]. It should be understood that the at least one processor module [202] may consist of any number of devices. The at least one processor module [202] may include a data processing device, such as a microprocessor or microcontroller or a central processing unit. The at least one processor module [202] may include another logic device such as a DMA (Direct Memory Access) processor, an integrated communication processor device, a custom VLSI (Very Large Scale Integration) device or an ASIC (Application Specific Integrated Circuit) device. In addition, the at least one processor module [202] may include any other type of analog or digital circuitry that is designed to perform the processing functions described herein.

As shown in FIG. 2, in computer system [200], the at least one memory module [204], the at least one processor module [202], and secondary memory module [218] are all operably linked together through communication infrastructure [220], which may be a communications bus, system board, cross-bar, etc.). Through the communication infrastructure [220], computer program codes or instructions or computer readable data can be transferred and exchanged. Input interface may operably connect the at least one input module [226] to the communication infrastructure [220]. Likewise, output interface [222] may operably connect the at least one output module [224] to the communication infrastructure [220].

The at least one input module [230] may include, for example, a keyboard, mouse, touch screen, scanner, and other input devices known in the art. The at least one output module [224] may include, for example, a display screen, such as a computer monitor, TV monitor, or the touch screen of the at least one input module [230]; a printer; and audio speakers. Computer system [200] may also include, modems, communication ports, network cards such as Ethernet cards, and newly developed devices for accessing intranets or the internet.

The at least one memory module [206] may be configured for storing patient data entered via the at least one input module [230] and processed via the at least one processor module [202]. Patient data relevant to the present invention may include sequence information for one or more of the genes in any of Panels A-Q. Patient data relevant to the present invention may also include clinical parameters relevant to the patient (e.g., age, lifestyle and environmental risk factors for cancer, previously diagnosed diseases (including previously diagnosed cancers), tumor size, node status, tumor stage). Any patient data a physician might find useful in making treatment decisions/recommendations may also be entered into the system, including but not limited to age, gender, and race/ethnicity and lifestyle data such as diet information. Other possible types of patient data include

symptoms currently or previously experienced, patient's history of illnesses, medications, and medical procedures.

The at least one memory module [206] may include a computer-implemented method stored therein. The at least one processor module [202] may be used to execute software or computer-readable instruction codes of the computer-implemented method. The computer-implemented method may be configured to, based upon the patient data, indicate whether the patient has an increased likelihood of recurrence, progression or response to any particular treatment, generate a list of possible treatments, etc.

In certain embodiments, the computer-implemented method may be configured to identify a patient as having or not having an increased risk of a particular cancer. For example, the computer-implemented method may be configured to inform a physician that a particular patient has an increased risk of a particular cancer. Alternatively or additionally, the computer-implemented method may be configured to actually suggest a particular course of treatment based on the answers to/results for various queries.

FIG. 3 illustrates one embodiment of a computer-implemented method [300] of the invention that may be implemented with the computer system [200] of the invention. The method begins with one of multiple queries ([310], [311], [312]), either sequentially or substantially simultaneously. If the answer to/result for any of these queries is "Yes" [320], the method concludes that the patient has an increased risk of a particular cancer (e.g., breast cancer if there is a germline deficiency in BRCA1). If the answer to/result for all of these queries is "No" [321], the method concludes [331] that the patient does not have, at least based on germline status of the tested genes, an increased risk of cancer. The method [300] may then proceed with more queries, make a particular treatment recommendation ([340], [341]), or simply end.

When the queries are performed sequentially, they may be made in the order suggested by FIG. 3 or in any other order. Whether subsequent queries are made can also be dependent on the results/answers for preceding queries. In some embodiments of the method illustrated in FIG. 3, for example, the method asks about BRCA1 [311] first and, if the patient has a germline deficiency then the method concludes [330] or optionally confirms by BRCA2 status [311], and/or other HCG status [312]. Optionally, the method may query clinical parameters (e.g., tumor size, age, tumor stage) before or after querying any of the molecular characteristics of HCGs as shown. As mentioned above, the preceding order of queries may be modified. In some embodiments an answer of "yes" to one query (e.g., [310]) prompts one or more of the remaining queries to confirm that the patient has, e.g., increased risk of recurrence.

In some embodiments, the computer-implemented method of the invention [300] is open-ended. In other words, the apparent first step [310] in FIG. 3 may actually form part of a larger process and, within this larger process, need not be the first step/query. Additional steps may also be added onto the core methods discussed above. These additional steps include, but are not limited to, informing a health care professional (or the patient itself) of the conclusion reached; combining the conclusion reached by the illustrated method [300] with other facts or conclusions to reach some additional or refined conclusion regarding the patient's diagnosis, prognosis, treatment, etc.; making a recommendation for treatment (e.g., "patient should/should not undergo prophylactic mastectomy"); additional queries about additional biomarkers, clinical parameters (e.g., age, tumor size, node

status, tumor stage), or other useful patient information (e.g., age at diagnosis, general patient health, etc.).

Regarding the above computer-implemented method [300], the answers to the queries may be determined by the method instituting a search of patient data for the answer. For example, to answer the respective queries ([310], [311], [312]), patient data may be searched for germline sequence data for the HCGs to be analyzed (e.g., the genes in Panel B). The queries may be performed in no particular order or according to some desired order (e.g., in order of gene number in Panel B). If such a comparison has not already been performed, the method may compare these data to some reference in order to determine if the patient has a germline deficiency in any of the HCGs being analyzed. Additionally or alternatively, the method may present one or more of the queries ([310], [311], [312]) to a user (e.g., a physician) of the computer system [200]. For example, the questions ([310], [311], [312]) may be presented via an output module [224]. The user may then answer "Yes" or "No" or provide some other value (e.g., numerical or qualitative value representing germline HCG status) via an input module [230]. The method may then proceed based upon the answer received. Likewise, the conclusions [330, 331] may be presented to a user of the computer-implemented method via an output module [224].

The practice of the present invention may also employ conventional biology methods, software and systems. Computer software products of the invention typically include computer readable media having computer-executable instructions for performing the logic steps of the method of the invention. Suitable computer readable medium include floppy disk, CD-ROM/DVD/DVD-ROM, hard-disk drive, flash memory, ROM/RAM, magnetic tapes and etc. Basic computational biology methods are described in, for example, Setubal et al., *INTRODUCTION TO COMPUTATIONAL BIOLOGY METHODS* (PWS Publishing Company, Boston, 1997); Salzberg et al. (Ed.), *COMPUTATIONAL METHODS IN MOLECULAR BIOLOGY*, (Elsevier, Amsterdam, 1998); Rashidi & Buehler, *BIOINFORMATICS BASICS: APPLICATION IN BIOLOGICAL SCIENCE AND MEDICINE* (CRC Press, London, 2000); and Ouelette & Bzevanis, *BIOINFORMATICS: A PRACTICAL GUIDE FOR ANALYSIS OF GENE AND PROTEINS* (Wiley & Sons, Inc., 2nd ed., 2001); see also, U.S. Pat. No. 6,420,108.

The present invention may also make use of various computer program products and software for a variety of purposes, such as probe design, management of data, analysis, and instrument operation. See U.S. Pat. Nos. 5,593,839; 5,795,716; 5,733,729; 5,974,164; 6,066,454; 6,090,555; 6,185,561; 6,188,783; 6,223,127; 6,229,911 and 6,308,170. Additionally, the present invention may have embodiments that include methods for providing genetic information over networks such as the Internet as shown in U.S. Ser. No. 10/197,621 (U.S. Pub. No. 20030097222); U.S. Ser. No. 10/063,559 (U.S. Pub. No. 20020183936); U.S. Ser. No. 10/065,856 (U.S. Pub. No. 20030100995); U.S. Ser. No. 10/065,868 (U.S. Pub. No. 20030120432); U.S. Ser. No. 10/423,403 (U.S. Pub. No. 20040049354).

The terms "probe" and "oligonucleotide" (also "oligo"), when used in the context of nucleic acids, interchangeably refer to a relatively short nucleic acid fragment or sequence. The invention also provides primers useful in the methods of the invention. "Primers" are oligonucleotides capable, under the right conditions and with the right companion reagents, of selectively amplifying a target nucleic acid (e.g., a target exon or gene). In the context of nucleic acids, "probe" is used herein to encompass "primer" since primers can generally also serve as probes.

The probe can generally be of any suitable size/length. In some embodiments the probe is between A and B nucleotides in length. In some embodiments A=10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, or 90,000, or more and B=15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, 90,000, or 100,000 or more. These embodiments include every combination of A and B as set forth in the preceding sentence, where B>A. For example, the probe may comprise (or consist of or consist essentially of) a range of nucleotides in length from any A to any B (e.g., from 10 to 15, 10 to 20, [. . .] 100 to 125, 100 to 150, etc.). In some embodiments the probe has a length from about 8 to 200, 15 to 150, 15 to 100, 15 to 75, 15 to 60, or 20 to 55 bases in length. They can be labeled with detectable markers with any suitable detection marker including but not limited to, radioactive isotopes, fluorophores, biotin, enzymes (e.g., alkaline phosphatase), enzyme substrates, ligands and antibodies, etc. See Jablonski et al., *NUCLEIC ACIDS RES.* (1986) 14:6115-6128; Nguyen et al., *BIOTECHNIQUES* (1992) 13:116-123; Rigby et al., *J. MOL. BIOL.* (1977) 113:237-251. Indeed, probes may be modified in any conventional manner for various molecular biological applications. Techniques for producing and using such oligonucleotide probes are conventional in the art.

Probes according to the invention can be used in the hybridization, amplification, detection or sequencing techniques discussed above. Thus, some embodiments of the invention comprise probe sets (including primer sets) suitable for use in detecting, amplifying, quantitating, and/or sequencing HCGs of the invention. In some embodiments the probe sets have a certain proportion of their probes directed to HCGs (e.g., HCGs in any of Panels A-Q)—e.g., a probe set consisting of 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% probes specific for HCGs.

The total number of genes to which the probes in the probe set are directed may vary depending on resource and technical constraints. In some embodiments the probe set comprises (or consists of or consists essentially of) probes directed to between W and X genes, where W=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, or 55 or more and X=3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, or 20,000 or more. These embodiments include every combination of W and X as set forth in the preceding sentence, where X>W. For example, the plurality of genes to which probes in the probes set are directed may comprise (or consist of or consist essentially of) a range of genes in number from any W to any X (e.g., from 10 to 15, 10 to 20, [. . .] 100 to 125, 100 to 150, etc.).

In some embodiments the genes to which probes in the probe set are directed will comprise at least some of the genes listed in Panels A-Q. Thus, in some embodiments the probe set comprises probes directed to at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, or 55 genes listed in Panels A-Q. In some embodiments the probe set comprises probes directed to between Y and Z gene of any of Panels A-Q, wherein Y=1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, or 54 and Z=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55. In some embodiments, said plurality of genes comprises gene numbers 1 & 2, 2 & 3, 3 & 4, 4 & 5, 5 & 6, 6 & 7, 7 & 8, 8 & 9, 9 & 10, 10 & 11, 11 & 12, 12 & 13, 13 & 14, 14 & 15, 15 & 16, 16 & 17, 17 & 18, 18 & 19, 19 & 20, 20 & 21, 21 & 22, 22 & 23, 23 & 24, 24 & 25, 25 & 26, 26 & 27, 27 & 28, 28 & 29, 29 & 30, 30 & 31, 31 & 32, 32 & 33, 33 & 34, 34 & 35, 35 & 36, 36 & 37, 37 & 38, 38 & 39, 39 & 40, 40 & 41, 41 & 42, 42 & 43, 43 & 44, 44 & 45, 45 & 46, 46 & 47, 47 & 48, 48 & 49, 49 & 50, 50 & 51, 51 & 52, 52 & 53, 53 & 54, or 54 & 55 from any of Panels A-Q. These embodiments include every combination of Y and Z as set forth in the preceding sentences, where Y>Z. For example, the probe set comprises (or consists of or consists essentially of) probes directed to a range of genes with a number from any Y to any Z in any of Panels A-Q (e.g., from 1 to 2, 1 to 3, 1 to 4, [. . .] 1 to 55, 2 to 3, 2 to 4, 2 to 5, [. . .] 2 to 55, etc.).

As used herein, a probe (or primer) is "directed to" a gene when such probe hybridizes under some minimal stringency conditions (e.g., high stringency conditions) to a nucleic acid comprising a nucleotide sequence specific for such gene (e.g., in the genome essentially only found in that gene).

In another aspect of the present invention, a kit is provided for practicing the prognosis of the present invention. The kit may include a carrier for the various components of the kit. The carrier can be a container or support, in the form of, e.g., bag, box, tube, rack, and is optionally compartmentalized. The carrier may define an enclosed confinement for safety purposes during shipment and storage. The kit may include oligonucleotides directed to (e.g., specifically hybridizing under high stringency to) mRNA or cDNA of a plurality of genes in any of Panels A-Q. Such oligonucleotides can be used as PCR primers in RT-PCR reactions, or hybridization probes. In some embodiments the kit comprises reagents (e.g., probes, primers, and/or antibodies) for determining the sequence of a panel of genes, where said panel comprises at least 25%, 30%, 40%, 50%, 60%, 75%, 80%, 90%, 95%, 99%, or 100% HCGs (e.g., HCGs in any of Panels A-Q). In some embodiments the kit consists of reagents (e.g., probes, primers, and/or antibodies) for determining the expression level of no more than 2500 genes, wherein at least 5, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100, 120, 150, 200, 250, or more of these genes are HCGs (e.g., HCGs in any of Panels A-Q).

The oligonucleotides in the detection kit can be labeled with any suitable detection marker including but not limited to, radioactive isotopes, fluorophores, biotin, enzymes (e.g., alkaline phosphatase), enzyme substrates, ligands and antibodies, etc. See Jablonski et al., *NUCLEIC ACIDS RES.*, 14:6115-6128 (1986); Nguyen et al., *BIOTECHNIQUES*, 13:116-123 (1992); Rigby et al., *J. MOL. BIOL.*, 113:237-251 (1977).

Alternatively, the oligonucleotides included in the kit are not labeled, and instead, one or more markers are provided in the kit so that users may label the oligonucleotides at the time of use.

Various other components useful in the detection techniques may also be included in the detection kit of this invention. Examples of such components include, but are not limited to, Taq polymerase, deoxyribonucleotides, dideoxyribonucleotides, other primers suitable for the amplification of a target DNA sequence, RNase A, and the like. In addition, the detection kit preferably includes instructions on using the kit for practice the prognosis method of the present invention using human samples.

EXAMPLE 1

Biological samples from patients that can yield germline DNA are obtained. Genomic DNA is extracted from biological samples, purified, and quantitated. Genomic regions of interest (i.e., exons of the genes of interest plus on average 10 flanking intronic nucleotides on each side of each exon) are enriched by amplification using primers specific for these regions. Genes analyzed in this example are those of Panel F.

Genomic DNA is fragmented and subjected to a merge on a RainDance instrument with a target enrichment PCR primer library. The library is designed to amplify approximately 1,200 targets covering all coding regions (plus on average 10 flanking intronic nucleotides on each side of each exon) of the genes in Panel F. Specifically, one micro-droplet at a time, the merging process melds together in an oil phase a micro-droplet containing one or more DNA fragments from the patient sample (or derived, e.g., amplified, therefrom) with a micro-droplet containing thousands of copies of one or more primer pairs targeting widely-spaced unique positions of interest (this example involves 5 primer pairs as one preferred embodiment, but 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 or more primer pairs may be used within a droplet). The process is repeated approximately from 1 to 2 million times. The collection of merged droplets is subjected to emulsion PCR amplification. The emulsion is disrupted, cleaned up, and subjected to secondary PCR that tails the primary PCR products with sequencing primers, anchors and an indexing barcode for the Illumina sequencing process. Samples from one or more patients are pooled together for sequencing (this example involves pooling of samples from 96 patients, but samples from 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 192, 200, 225, 250, 275, 300 or more patients may be pooled).

Some genes (e.g., PMS2, CHEK2) encompass genomic areas with pseudogenes. Pseudogenes may interfere with normal sequencing. For those genes, genomic DNA is also amplified with gene-specific primers to produce long range PCR products. The long range PCR products are used as surrogate gene targets for sequencing. Specifically, the long range products are amplified with a 4-primer PCR mix containing Illumina adapter-tailed primary nested primer sets specific to the genes, as well as secondary primers containing sequencing chip anchor sequences, indexing barcodes and designed to prime off the Illumina adapter tails of the primary primers.

Amplified DNA is sequenced using the Illumina MiSeq™ (or analogous Hi-Seq™) system according to the manufacturer's protocol. This system yields high quality sequence data for each exon amplified.

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Sequence data are compared to reference sequences using alignment software to determine whether each patient has a germline variation in any of the genes of interest. Further analysis is performed to determine whether any such variation is deleterious, including looking for nonsense and frame-shift variants or large rearrangements.

EXAMPLE 2

This Example 2 describes a study performed to assess a panel of the invention in a large population of patients suspected of having hereditary breast and ovarian cancer syndrome (HBOC), e.g., patients suspected of having a BRCA1 and/or BRCA2 mutation. The details of DNA preparation and sequencing were as described in Example 1 above, except Panel B was assessed instead of Panel F. DNA from 1955 prospectively accrued cases was anonymized for this study. Patients with Ashkenazi Jewish heritage were excluded in order to determine the relative prevalence of mutations in a generalizable population. Extracted genomic DNA from blood was hybridized with a custom amplicon library on a Raindance™ ThunderStorm™ instrument. DNA was sequenced on an Illumina™ HiSeq2500™ system. Sequence variations, large rearrangements and large deletions among the 25 genes of Panel B were detected.

A total of 275/1955 (14.07%) patients were found to be mutation carriers in at least one of the genes of Panel B. 182/1955 (9.31%) patients had a mutation in BRCA1 or BRCA2. 96/1955 (4.91%) patients had a mutation in other genes. The distribution by gene of 96 probands with other gene mutations is shown in Table A below. The genes of Table A form yet another panel of the invention (Panel P) and these genes, together with the BRCA1 and BRCA2 genes, form Panel D.

TABLE A

Panel P			
Gene #	Gene Symbol	# patients with mutation	(%)
1	CHEK2	30	31.25%
2	ATM	14	14.58%
3	NBN	14	14.58%
4	PALB2	13	13.54%
5	BARD1	7	7.29%
6	BRIP1	7	7.29%
7	PMS2	4	4.17%
8	MSH2	2	2.08%
9	MSH6	2	2.08%
10	TP53	2	2.08%
11	MUTYH	1	1.04%

1738/1955 patients had a personal history of breast cancer. In 1091/1738 the incidence of breast cancer occurred prior to age 50, in 647/1738 the incidence of breast cancer occurred at or after age 50. Mutation prevalence for patients with breast cancer only, ovarian cancer only, breast and ovarian cancer or other HBOC cancers is shown in Table B below. 1902 of 1955 (97.29%) patients had a variant of uncertain clinical significance (VUS) in at least one of the genes tested with a median of three VUSs per patient.

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TABLE B

Patient Cancer History	Patients (n)	Mutation Carriers	BRCA1/BRCA2	Other Panel B Gene
Breast CA < 50 years	1091	167 (15.31%)	116* (10.63%)	51 (4.67%)
Breast CA ≥ 50 years	647	70 (10.82%)	40** (6.18%)	30 (4.64%)
Ovarian CA	162	23 (14.20%)	17 (10.49%)	6 (3.70%)
Breast and Ovarian CA	40	12 (30.00%)	8 (20.00%)	4 (10.00%)
Other HBOC Cancer	15	3 (20.00%)	1 (6.67%)	2 (13.33%)

*2 and **1 patients had an additional mutation in a non-BRCA1/2 gene.

Panel B (more specifically Panel D) increased clinical sensitivity by 4.76% (95% C.I., 2.71-6.81%) in this study sample of 1955 patients as compared to BRCA1/BRCA2 testing alone. The observed improvement in clinical sensitivity achieved over BRCA1/BRCA2 testing alone is 51.1%. Thus, among cancer patients at risk for HBOC, Panel B (more specifically Panel D) results in a greater than 50% increase in mutation detection over current BRCA1/BRCA2 clinical testing. Panel P and preferably Panel D (or subpanels comprising the top 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 genes thereof) can therefore be particularly useful in targeted assessment of cancer risk in patients at risk of having HBOC.

EXAMPLE 3

This Example 3 describes a study performed to assess a panel of the invention in a population of patients suspected of having Lynch syndrome, e.g., patients submitted for testing of mismatch repair (MMR) genes (MLH1, MSH2, MSH6, PMS2, and EPCAM) based on having an indicator of Lynch syndrome. The details of DNA preparation and sequencing were as described in Example 1 above, except Panel B was assessed instead of Panel F. DNA from 343 prospectively accrued cases was anonymized for this study. Extracted genomic DNA from blood was hybridized with a custom amplicon library on a Raindance™ ThunderStorm™ instrument. DNA was sequenced on an Illumina™ HiSeq2500™ system. Sequence variations, large rearrangements and large deletions among the 25 genes of Panel B were detected.

Out of 343 cases, 45 (13%) had a mutation in MLH1, MSH2, MSH6 or PMS2. Out of 298 cases negative for these genes, other deleterious mutations were found as shown in Table C. The genes of Panel Q can be added to the MMR genes to form Panel E of the invention.

TABLE C

Panel Q				
Gene #		Cases w/ deleterious mutation		MMR mutation negative cases with other gene mutation
				% of total patients
1	BRCA2	6	6	1.75%
2	BRCA1	3	3	0.87%
3	RAD50	3	2	0.58%
4	BRIP1	2	2	0.58%
5	CHEK2	2	2	0.58%
6	ATM	2	1	0.29%
7	BARD1	1	1	0.29%

TABLE C-continued

Panel Q				
Gene #	Gene Symbol	Cases w/ deleterious mutation	MMR mutation negative cases with other gene mutation	
			#	% of total patients
8	MUTYH Bi-Allelic	1	1	0.29%
	MUTYH Mono-Allelic	7	5	1.46%
	Total excluding MYH mono-allelic	20	18	5.25%

Panel E increased clinical sensitivity by 5.25% in this study sample of 343 patients as compared to MMR gene testing alone. The observed improvement in clinical sensitivity achieved over MMR gene testing alone is 40.4%. To better understand the contribution of BRCA1 and BRCA2 to these suspected Lynch syndrome patients, the type of cancer that was the indicator for Lynch syndrome testing in the nine BRCA1- or BRCA2-positive patients was analyzed. All nine patients had at least one indicator of Lynch syndrome. In four cases, distinct indicators for both Lynch syndrome and HBOC (i.e., indicators not shared between the syndromes) were present. In four other cases, only indicators for Lynch syndrome were present. In one case, only a shared indicator for both Lynch syndrome and HBOC (i.e., ovarian cancer) was present. Even excluding this ovarian cancer case, BRCA2 and BRCA1 alone out of Panel E increased sensitivity by 2.33% over testing only the MMR genes. This translates to an observed improvement in clinical sensitivity over MMR gene testing alone of 17.9%. Thus, among cancer patients at risk for Lynch syndrome, Panel E results in a 40% increase in mutation detection over current MMR gene testing alone. Panel Q and preferably Panel E (or subpanels comprising the top 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 genes thereof) can therefore be particularly useful in targeted assessment of cancer risk in patients at risk of having Lynch syndrome.

Additional Embodiments

Embodiment 1. A method for sequencing nucleic acids comprising: (1) isolating a plurality of nucleic acid molecules from a sample taken from a patient, each nucleic acid molecule comprising between A and B nucleotides in length, said plurality of nucleic acid molecules comprising one or more exons of a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q; and (2) determining the sequence of said plurality of nucleic acid molecules.

Embodiment 2. A method for determining whether a patient has an increased risk of cancer, which comprises: (1) determining for a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q, whether the patient has a germline deficiency in any genes in said plurality of genes; and either (2) correlating a germline deficiency in any of said plurality of genes to an increased risk of cancer, or (3) correlating the absence of a germline deficiency in all of said plurality of genes to no increased risk of cancer.

Embodiment 3. The method of Embodiment 2 further comprising (a) isolating a plurality of nucleic acid molecules from a sample taken from a patient, each nucleic acid molecule comprising between A and B nucleotides in length,

and said plurality of nucleic acid molecules comprising one or more exons of said plurality of genes and (b) determining the sequence of said plurality of nucleic acid molecules.

Embodiment 4. The method of Embodiment 3, further comprising detecting a germline deficiency in a gene by comparing the sequence determined in (b) with one or more reference sequences.

Embodiment 5. A method for treating a patient comprising (1) determining for a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q, whether the patient has a germline deficiency in any genes in said plurality of genes; and (2)(a) correlating a germline deficiency in any of said plurality of genes to an increased risk of cancer, or (2)(b) correlating the absence of a germline deficiency in all of said plurality of genes to no increased risk of cancer; and (3) recommending, prescribing, or administering a treatment to reduce the patient's risk of cancer.

Embodiment 6. The method of Embodiment 5, wherein said treatment comprises surgery to remove all or part of the organ in which the patient has an increased risk of cancer.

Embodiment 7. The method of Embodiment 6, wherein said surgery is chosen from the group consisting of mastectomy, salpingo-oophorectomy, hysterectomy, colectomy, and prostatectomy.

Embodiment 8. The method of Embodiment 5, wherein said treatment comprises preventive drug treatment.

Embodiment 9. The method of Embodiment 8, wherein said preventive drug treatment comprises tamoxifen treatment.

Embodiment 10. A system comprising (1) computer program for receiving, storing, and/or retrieving a patient's sequence data for a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; (2) computer program for querying this patient data; (3) optionally a computer program for comparing the patient's sequence data to one or more reference sequences to determine whether there is a mutation; (4) computer program for concluding whether there is an increased likelihood of cancer based on the presence or absence of a mutation; and optionally (4) computer program for outputting/displaying this conclusion.

Embodiment 11. A system for sequencing genes in a sample, comprising: (1) a sample analyzer for sequencing a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q, wherein the sample analyzer contains (a) the sample which is from a patient, (b) genomic DNA from the sample, (c) transcript RNA from the sample, or (d) DNA synthesized from said genomic DNA; (2) a first computer program for receiving test sequence data on the plurality of genes; and (3) a second computer program for comparing the sequence data to one or more reference sequences.

Embodiment 12. The system of Embodiment 11, comprising a computer program for determining the patient's degree of risk of cancer based at least in part on the comparison of the test sequence with said one or more reference sequences.

Embodiment 13. The system of Embodiment 12, wherein said computer program for determining the patient's degree of risk of cancer compares the patient's determined probability of a particular cancer with a reference probability to determine whether the patient has an increased risk of such cancer.

Embodiment 14. A composition comprising:
(a) nucleic acid probes hybridizing to a plurality of nucleic acid molecules comprising one or more exons

of a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q;

- (b) nucleic acid primers and primer pairs suitable for selectively amplifying nucleic acids of (a);
- (c) antibodies binding immunologically to polypeptides encoded by a plurality of genes consisting of between W and X genes, and said plurality of genes comprising at least two genes in any of Panels A-Q;
- (d) a probe set comprising (a), (b) and/or (c); or
- (e) a microarray comprising (a), (b), (c), and/or (d).

Embodiment 15. A kit comprising: reagents for sequencing nucleic acid molecules comprising one or more exons of a plurality of genes comprising a plurality of genes consisting of between W and X genes, said plurality of genes comprising at least two genes in any of Panels A-Q; and instructions for using said reagents.

Embodiment 16. The kit of Embodiment 15, comprising a composition of claim 14.

Embodiment 17. The kit of Embodiment 15, wherein said reagents are PCR primers specific for the plurality of genes.

Embodiment 18. The kit of Embodiment 15, wherein said reagents are PCR primers specific for the exons of the plurality of genes.

Embodiment 19. The kit of Embodiment 15, wherein said reagents are oligonucleotide probes specific for the exons of the plurality of genes.

Embodiment 20. The kit of Embodiment 15, wherein said reagents are packaged into an array.

Embodiment 21. The method of any one of Embodiments 1, 3, or 4, comprising comparing the sequences determined in an earlier step with one or more reference sequences.

Embodiment 22. The method of Embodiment 21, comprising correlating a difference between the determined sequences and the one or more reference sequences to a mutation in one or more of the genes in the plurality of genes.

Embodiment 23. The method of Embodiment 21 or Embodiment 22, wherein the reference sequence for any given gene in the plurality is any of the sequences corresponding to that gene as shown in Table 3.

Embodiment 24. The system of any one of Embodiments 10-13, comprising a computer program for determining whether the patient has a mutation in one or more of the genes in the plurality of genes by determining whether there is a difference between the determined sequences and the one or more reference sequences.

Embodiment 25. The system of Embodiment 24, wherein the reference sequence for any given gene in the panel is any of the sequences corresponding to that gene as shown in Table 3.

Embodiment 26. The method of any one of Embodiments 1-9, or 21-23, comprising correlating a germline deficiency in any particular gene in the plurality of genes to an increased risk of a particular cancer as shown in Table 4.

Embodiment 27. The method of any one of Embodiments 1-9, 21-23, or 26, comprising diagnosing the patient with an increased risk of a particular cancer as shown in Table 4 based at least in part on a germline deficiency in any particular gene in the plurality of genes.

Embodiment 28. The method of any one of Embodiments 1-9, 21-23, comprising correlating no germline deficiency in any gene in the plurality of genes with no increased risk of any cancer.

Embodiment 29. The system of any one of Embodiments 10-13, comprising a computer program for determining the patient's degree of risk of any particular cancer as shown in

Table 4 based at least in part on the comparison of the test sequence with said one or more reference sequences.

Embodiment 30. The method of any of Embodiments 1, 3 or 4, wherein A=10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, or 90,000, or more; and B=15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, 20,000, 25,000, 30,000, 35,000, 40,000, 45,000, 50,000, 60,000, 70,000, 80,000, 90,000, or 100,000 or more.

Embodiment 31. The method of any of Embodiments 1, 3 or 4, wherein said plurality of DNA molecules comprises at least some length of intronic sequence adjacent to at least one of said one or more exons.

Embodiment 32. The method of Embodiment 31, wherein said plurality of DNA molecules comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 200, 300, 400, 500 or more base pairs of the intronic sequence on one or both sides of the at least one exon.

Embodiment 33. The method of any one of Embodiments 1-10, 21-23, 26-28, or 30-32, wherein W=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 or more; and X=3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, or 20,000 or more.

Embodiment 34. The system of any one of Embodiments 10-13, 24, 25, or 29, wherein W=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 or more; and X=3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 5,000, 6,000, 7,000, 8,000, 9,000, 10,000, 12,000, 14,000, 16,000, 18,000, or 20,000 or more.

Embodiment 35. The method of any one of Embodiments 1-10, 21-23, 26-28, or 30-33, wherein said plurality of genes comprises at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 genes listed in any of Panels A-Q.

Embodiment 36. The system of any one of Embodiments 10-13, 24, 25, 29, or 34, wherein said plurality of genes comprises at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 genes listed in any of Panels A-Q.

Embodiment 37. The method of any one of Embodiments 1-10, 21-23, 26-28, 30-33, or 35, wherein the plurality of genes comprises gene numbers between Y and Z of any of

Panels A-Q and Y=1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67 or 68 and Z=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69.

Embodiment 38. The method of any one of Embodiments 1-10, 21-23, 26-28, 30-33, 35, or 37, wherein said plurality of genes comprises gene numbers 1 & 2, 2 & 3, 3 & 4, 4 & 5, 5 & 6, 6 & 7, 7 & 8, 8 & 9, 9 & 10, 10 & 11, 11 & 12, 12 & 13, 13 & 14, 14 & 15, 15 & 16, 16 & 17, 17 & 18, 18 & 19, 19 & 20, 20 & 21, 21 & 22, 22 & 23, 23 & 24, 24 & 25, 25 & 26, 26 & 27, 27 & 28, 28 & 29, 29 & 30, 30 & 31, 31 & 32, 32 & 33, 33 & 34, 34 & 35, 35 & 36, 36 & 37, 37 & 38, 38 & 39, 39 & 40, 40 & 41, 41 & 42, 42 & 43, 43 & 44, 44 & 45, 45 & 46, 46 & 47, 47 & 48, 48 & 49, 49 & 50, 50 & 51, 51 & 52, 52 & 53, 53 & 54, 54 & 55, 55 & 56, 56 & 57, 57 & 58, 58 & 59, 59 & 60, 60 & 61, 61 & 62, 62 & 63, 63 & 64, 64 & 65, 65 & 66, 66 & 67, 67 & 68, or 68 & 69 of any of Panels A-Q.

Embodiment 39. The method of any one of Embodiments 1-10, 21-23, 26-28, 30-33, 35, or 37-38, wherein the genes chosen from Panels A-Q comprise at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%, of the plurality of genes to be analyzed.

Embodiment 40. The system of any one of Embodiments 10-13, 24, 25, 29, 34, or 36, wherein said plurality of genes comprises at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69 genes listed in any of Panels A-Q.

Embodiment 41. The system of any one of Embodiments 10-13, 24, 25, 29, 34, 36, or 40, wherein the plurality of genes comprises gene numbers between Y and Z of any of Panels A-Q and Y=1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30,

31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67 or 68 and Z=2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, or 69.

Embodiment 42. The system of any one of Embodiments 10-13, 24, 25, 29, 34, 36, or 40-41, wherein said plurality of genes comprises gene numbers 1 & 2, 2 & 3, 3 & 4, 4 & 5, 5 & 6, 6 & 7, 7 & 8, 8 & 9, 9 & 10, 10 & 11, 11 & 12, 12 & 13, 13 & 14, 14 & 15, 15 & 16, 16 & 17, 17 & 18, 18 & 19, 19 & 20, 20 & 21, 21 & 22, 22 & 23, 23 & 24, 24 & 25, 25 & 26, 26 & 27, 27 & 28, 28 & 29, 29 & 30, 30 & 31, 31 & 32, 32 & 33, 33 & 34, 34 & 35, 35 & 36, 36 & 37, 37 & 38, 38 & 39, 39 & 40, 40 & 41, 41 & 42, 42 & 43, 43 & 44, 44 & 45, 45 & 46, 46 & 47, 47 & 48, 48 & 49, 49 & 50, 50 & 51, 51 & 52, 52 & 53, 53 & 54, 54 & 55, 55 & 56, 56 & 57, 57 & 58, 58 & 59, 59 & 60, 60 & 61, 61 & 62, 62 & 63, 63 & 64, 64 & 65, 65 & 66, 66 & 67, 67 & 68, or 68 & 69 of any of Panels A-Q.

Embodiment 43. The system of any one of Embodiments 10-13, 24, 25, 29, 34, 36, or 40-42, wherein the genes chosen from Panels A-Q comprise at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%, of the plurality of genes to be analyzed.

All publications and patent applications mentioned in the specification are indicative of the level of those skilled in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference. The mere mentioning of the publications and patent applications does not necessarily constitute an admission that they are prior art to the instant application.

Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be obvious that certain changes and modifications may be practiced within the scope of the appended claims.

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12410478B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A method comprising:

- a) extracting genomic DNA from a sample comprising germline cells of a patient;
- b) forming a mixture by hybridizing nucleic acid probes that hybridize to the genomic DNA, wherein the genomic DNA comprises a plurality of nucleic acid molecules, wherein said plurality of nucleic acid molecules comprise the coding regions of a plurality of test genes and wherein said plurality of test genes comprise

target genes: APC gene, ATM gene, BARD 1 gene, BMPRIA gene, Breast Cancer 1 (BRCA1) gene, Breast Cancer 2 (BRCA2) gene, BRIP1 gene, CDH1 gene, CDK4 gene, CDKN2A gene, CHEK2 gene, EPCAM gene, MLH1 gene, MSH2 gene, MSH6 gene, MUTYH gene, NBN1 gene, PALB2 gene, PMS2 gene, phosphatase and tensin homolog (PTEN) gene, RADSIC gene, RADS1D gene, SMAD4 gene, STK11 gene, and TP53 gene;

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- c) performing one or more amplification reactions for amplifying said plurality of nucleic acid molecules in the DNA mixture in the presence of one or more primer pairs, wherein each of the primer pairs generates amplification products comprising a portion of each of the target genes;
- d) sequencing said amplification products; and
- e) comparing the sequences of the amplification products with one or more reference sequences of each of the target genes using an alignment software, thereby determining whether the patient has germline mutations in the target genes based on the presence of at least one insertion or deletion greater than 1,000 nucleotides in the sequences of the amplification products from the plurality of test genes.
2. The method of claim 1, wherein the one or more reference sequences comprise the sequence of each of the target genes in the following table:

Entrez Gene Symbol	SEQ ID NO
APC	1
ATM	20
BARD1	21
BMPR1A	60
BRCA1	97
BRCA2	128
BRIP1	158
CDH1	182
CDK4	201
CDKN2A	212
CHEK2	223
EPCAM	244
MLH1	289
MSH2	355
MSH6	374
MUTYH	387
NBN	411
PALB2	430
PMS2	450
PTEN	475

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-continued

Entrez Gene Symbol	SEQ ID NO
RAD51C	516
RAD51D	521
SMAD4	546
STK11	561
TP53	576.

3. The method of claim 1, wherein at least one of the amplification products comprises at least a portion of an exon sequence and an intron sequence comprising at least 10 base pairs in length flanking at least one end of said exon sequence.

4. The method of claim 1 further comprising determining if the patient has at least one second level risk factors comprising personal risk factors and family risk factors.

5. The method of claim 4, wherein the personal risk factors comprise that the patient has a history of multiple primary cancers, a positive triple negative breast cancer, a history of ovarian cancer, a history of smoking, a positive tissue biopsy for a cancer, a positive vaginal pap smear for a cancer, a history of male breast cancer, an enlarged prostate, colon polyps, and an age of the patient if a cancer is diagnosed, an age of menopause of the patient, and an age of menarche of the patient.

6. The method of claim 4, wherein the family risk factors comprise that the patient has Ashkenazi Jewish ancestry, a relative with early onset cancer, a relative with multiple primary cancers, a relative with male breast cancer, a relative with ovarian cancer, and a relative with triple negative breast cancer.

7. The method of claim 1, wherein the target genes comprise at least 25% of the plurality of test genes.

8. The method of claim 1, wherein the target genes comprise at least 50% of the plurality of test genes.

9. The method of claim 1, wherein the sample is a human sample.

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